# Who Holds U.S. Bonds and How It Shapes the Yield Effects of QE and QT

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**Abstract**

How much do the bond-supply shifts of quantitative easing (QE) and tightening (QT) move yields? The answer turns on the slope of aggregate bond demand: who holds bonds, how sensitive their demand is to yields, and how fast they rebalance. I estimate heterogeneous investor demand in the U.S. bond market across four asset classes (Treasuries, agency MBS, corporate bonds, and municipal bonds) from sector-level portfolio-share data spanning 1985 to 2025, using granular, QE-shock, and heteroskedasticity-based instruments. I measure each sector's demand against its own non-bond alternative, and I let each sector rebalance at its own speed. Most investors respond weakly to yields, and much of the response that does come arrives only over the following year. Because inelastic sectors hold most of the market, aggregate demand is inelastic. Feeding the demand system through market clearing, I find that historical QE compressed yields substantially in the markets the Fed bought, while the spillovers to corporate and municipal yields are an order of magnitude smaller: investors substitute mainly between bonds and non-bond assets, not across bond classes, so balance-sheet policy is narrow. QT's effects are smaller, because the float has since shifted toward the elastic sectors: a further \$1 trillion of runoff raises 10-year Treasury yields by about 24 basis points and agency MBS yields by about 18.

**Keywords:** Agency MBS, Demand System, Institutional Investors, Quantitative Easing, Quantitative Tightening, U.S. Treasuries

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# 1 Introduction

For over a decade, the Federal Reserve has purchased more than \$6 trillion of U.S. Treasuries and agency MBS to lower yields, a policy known as quantitative easing (QE). It has since reversed course, shrinking its balance sheet through quantitative tightening (QT), yet the consequences for bond yields and the channel through which they operate remain unclear.[1] The key is that QE and QT shift the supply of bonds available to private investors, so their effect on yields depends on the slope of investor demand curves: flat demand implies small yield effects, steep demand large ones (e.g., Koijen and Yogo, 2019; Vayanos and Vila, 2021). Because investors differ in their yield sensitivity, in the assets they hold, and in how quickly they rebalance, the aggregate demand curve, and hence the size and timing of the yield effect of QT, depends on *which* investors absorb the released supply, and how fast.

This paper estimates bond demand using sector-level portfolio data covering four asset classes, namely Treasuries, agency MBS, corporate bonds, and municipal bonds, across 26 sectors grouped into eight broad investor types: banks, insurers, pension funds, mutual funds, arbitrageurs, foreign investors, households, and other investors. In the demand system, each sector allocates wealth between the four bond classes and an outside asset, which I let differ across sectors. I measure each sector's bond demand against the non-bond class it actually rebalances toward, for example, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers. This sector-specific choice captures both substitution among bond classes and substitution between bonds and each holder's realistic alternative, so the demand system speaks to the full portfolio-rebalancing channel of QE and QT rather than to within-bond substitution alone. I let each sector rebalance at its own speed as well as toward its own outside asset, estimating how far and how fast each adjusts. The estimates reveal heterogeneous, and slow-moving, substitution patterns and support counterfactual analyses of QE and QT episodes, their effects on yields, and their cross-asset spillovers.

[1] In June 2026, Federal Reserve Chairman Kevin Warsh made balance-sheet policy one of five priorities for the conduct of monetary policy, charging a dedicated task force to weigh "the benefits and risks of the current ample reserves regime, and the composition of the Fed's balance sheet" (Warsh, 2026).

I find three main results. First, bond demand is heterogeneous and inelastic in aggregate. Banks, insurers, pension funds, and foreign investors are all weakly yield-sensitive; the household residual carries an elastic point estimate that is statistically indistinguishable from zero, an artifact of its residual construction; and the clearly elastic sectors, mutual funds and the dealer-led arbitrageurs, hold only about a fifth of the market, so the aggregate demand curve is steep. Sectors differ not only in how much they respond to yields but in how fast they rebalance. The largest long-term investors are also the slowest: insurers complete only a third of their rebalancing on impact and reach their positive long-run slope within a quarter to a year, the slow-moving capital of Duffie (2010) and the bond-market counterpart of the horizon-rising demand elasticities in equities (van der Beck, 2026); pension funds barely respond at any horizon; banks complete their adjustment within a quarter or two, and even the arbitrageurs, who respond at once, build to their full response over several quarters. Second, feeding the estimated demand system through market clearing, I find that historical QE compressed Treasury and agency MBS yields substantially, most strongly under the pandemic-era QE4, and that the transmission is narrow. Purchases spill over to corporate and municipal markets through portfolio rebalancing, the more so when the Fed buys multiple asset classes at once, but the spillovers are roughly one-tenth the size of the own-market effects (under QE1, 7 basis points for corporate and 5 for municipal yields, against 50 for the Treasuries and 119 for the agency MBS the Fed bought), because investors substitute mainly between bonds and their outside assets rather than across bond classes: balance-sheet policy concentrates its impact in the markets the Fed actually buys. Third, QE and QT are not mirror images: relative to its scale, QT moves yields by less, because the float has since shifted toward the elastic sectors, mutual funds and the arbitrageurs, so each unit of released supply meets a flatter demand curve. A further \$1 trillion of runoff, about a year and a half at the Fed's 2024 runoff pace of \$172 billion per quarter, would lift 10-year Treasury yields by about 24 basis points and agency MBS yields by about 18. Because much of the market's absorption capacity arrives with a lag, a hypothetical one-off balance-sheet change would move yields on impact by roughly three times its long-run effect, with the excess dissipating over the following year as slow capital and fund flows arrive; spread over the gradual pace of actual QT, the realized transient is far smaller, which is what quantifies the value of running QT gradually.

I construct quarterly sector-level portfolio shares for four U.S. bond asset classes (Treasuries, agency MBS, corporate bonds, and municipal bonds) from market-value balance sheets in the Federal Reserve's Flow of Funds (the Financial Accounts of the United States, Z.1). These shares form a quarterly panel structured by investor sector and asset class. Because the Flow of Funds reports holdings at market value, a change in a sector's share reflects both active trading and the mechanical revaluation of existing positions when yields move; I remove the revaluation with a duration-based valuation adjustment, so demand is identified from active rebalancing (Appendix D.3). I merge this panel with asset characteristics (yield, time to maturity, outstanding issuance, duration, and coupon) constructed from representative yield series (the 10-year Treasury yield; an option-adjusted agency-MBS yield, the 10-year Treasury yield plus the MBS index option-adjusted spread, which strips the prepayment-option premium from the index yield; Moody's Baa corporate yield; and the Bloomberg municipal index), the Bloomberg index characteristics, and the Flow of Funds, and with macroeconomic controls (GDP level, GDP growth, core PCE inflation, the broad dollar index, and the lagged VIX) drawn from FRED. The sample covers 26 sectors and four asset classes over the 161 quarters from 1985Q1 to 2025Q1, a span that reaches back to the high-yield years of the late 1980s. Those years are essential: they supply the yield variation that identifies the slow long-term investors, whose response is invisible at the near-zero yields prevailing after 2010.

Given these data, I estimate demand using a logit portfolio-share framework following Koijen and Yogo (2019). Each sector's demand follows a logit specification over the bond assets as a function of asset yields, characteristics, and idiosyncratic demand shocks. Four broad asset classes are also where the logit's proportional-substitution (IIA) restriction is least costly: substitution among close substitutes, securities of similar maturity within a class, is absorbed by the aggregation itself, so the restriction binds only across classes, where substitution is weak. I estimate the resulting demand equation in first differences, regressing the quarterly change in the log ratio of each bond's portfolio share to the outside-asset share on changes in asset yields and characteristics; differencing removes the secular comovement of yields and shares, and the valuation adjustment removes the mechanical revaluation of market-value holdings, so the yield coefficient is identified from active rebalancing rather than from trends or mark-to-market. Because sectors rebalance at different speeds, I trace the response horizon by horizon with local projections (Jordà, 2005), recovering each sector's impact slope, its long-run slope, and the speed of adjustment between them; the long-run slope is the demand parameter, whose yield coefficient governs the price elasticity. I recover these paths by projection rather than through the single inertia parameter with which van der Beck (2026) builds slow adjustment into an equity demand system, because any persistence in latent demand loads onto that parameter, conflating slow rebalancing with slowly moving preferences, and because some estimated paths, the mutual-fund sign flip among them, take shapes that geometric adjustment toward a fixed target cannot generate at any value of the parameter (Appendix D.6). The outside asset is sector-specific: I measure each sector's bond demand against the non-bond class it actually rebalances toward (for example, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers), so the estimated elasticity captures substitution between bonds and each holder's realistic alternative, directly reflecting the portfolio-rebalancing channel that QE aims to activate. Because the outside asset differs across sectors, the raw demand parameters are not directly comparable. The implied demand elasticity is, and it aggregates cleanly and governs the yield effects of QE and QT in the counterfactual analysis below.

Yields are endogenously determined by the interaction of supply and demand. I address this endogeneity with shifters of the supply each sector faces, unrelated to that sector's own demand, where they exist, and with heteroskedasticity-based instruments where they do not. For Treasuries I build a granular instrument, in the spirit of Gabaix and Koijen (2024), from the sectors' own portfolio data: the largest holders are big enough that their idiosyncratic rebalancing moves the market yield, and aggregating these idiosyncratic shocks, size-weighted and excluding the sector being instrumented, isolates supply-like variation orthogonal to that sector's own demand shock. Foreign demand corroborates its exogeneity: foreign Treasury holdings are reserve-driven (Warnock and Warnock, 2009), the foreign shock alone reproduces the granular estimate, and the joint specification passes the Sargan test. For agency MBS I use the Fed's own SOMA purchases, set by policy objectives rather than private MBS demand, together with the MBS-price surprise of Drukker (2025). For corporate and municipal bonds no relevant external supply shifter exists, their holders are diffuse, maturity walls are too small a share of yield variation, and foreign demand is absent or yield-chasing, so I use the heteroskedasticity-based instruments of Lewbel (2012).

The first-stage $F$ statistics are 12.1 for Treasuries, 11.0 for agency MBS, 7.3 for corporate, and 7.6 for municipal bonds. The Treasury and MBS blocks clear the conventional threshold and the heteroskedasticity blocks approach it; because none exceeds it comfortably, I base inference for the yield coefficient on weak-instrument-robust Anderson–Rubin confidence sets throughout (Anderson and Rubin, 1949; Andrews et al., 2019). The monetary-policy-surprise and Fed-purchase instruments used by the prior demand-system literature identify only the post-2010 sample and enter as robustness checks; the granular instrument identifies demand over the full sample.

The estimated demand parameters reveal substantial heterogeneity across sectors in the U.S. bond market. In the logit framework, a sector's demand elasticity scales with its yield sensitivity, its bond portfolio share, and the yield level, so the aggregate elasticity reflects not only how sensitive each sector is to yields but also who holds bonds. Banks, insurers, pension funds, and foreign investors are weakly yield-sensitive and, together with the household residual, whose elastic point estimate is statistically indistinguishable from zero, hold about four-fifths of bond assets; the elastic sectors, mutual funds and the arbitrageurs, hold the remaining fifth. Because the inelastic sectors dominate ownership, aggregate bond demand is inelastic: the estimated market elasticity ranges from about 0.3 over the QE era to 1.2 in the high-yield 1980s, consistent with the inelastic demand documented in the literature (e.g., Koijen and Yogo, 2019). The elasticity also varies with the horizon: the values above are long-run, and because the slow sectors barely rebalance on impact, aggregate demand is more inelastic still in the quarter of a shock, reaching these values only as the slower sectors complete their rebalancing over the following year, which is what makes a balance-sheet shock's initial yield impact overshoot its long-run level.

Lastly, I conduct a counterfactual analysis of Fed asset purchases using the estimated demand elasticities. Equilibrium yields are jointly determined by the demand system and the market-clearing condition that aggregate demand across sectors equals supply for each asset; together these define an implicit equilibrium yield as a function of supply, demand parameters, and asset characteristics. I evaluate the yield effect of the historical QE episodes by simulating counterfactual yields absent Fed purchases, holding demand parameters and characteristics fixed while letting bond supply respond to yields through reduced-form net-issuance elasticities estimated from the same Flow-of-Funds data; the fiscal path is held fixed throughout, so debt the Fed does not absorb or roll over is refinanced to the public and balance-sheet changes shift the private float one for one. Fed purchases lower the yields of purchased assets. Across the four programs, QE1 through the pandemic-era QE4, the impact is uniform once scaled: each 1 percent of the float removed compresses Treasury yields by roughly 9 to 12 basis points and agency MBS yields by 6 to 7. These magnitudes are comparable to other demand-system estimates (Figure 4); announcement-window event studies (e.g., Gagnon et al., 2011; Hancock and Passmore, 2011), which price the signaling channel as well as the stock effect but only the surprise component of purchases, are larger for QE1's Treasury effect and smaller for the later, largely anticipated programs. Purchases spill over to non-purchased markets through portfolio rebalancing, but narrowly: the corporate and municipal effects, 11 and 8 basis points under QE4, are an order of magnitude below the own-market effects, though larger when the Fed buys multiple bond types at once.

The same framework quantifies quantitative tightening, which reverses the sign of these effects. Per 1 percent of float returned to the market, the 2017–2019 and 2022–2025 runoffs raised Treasury yields by about 8 basis points, against the 9 to 12 that QE compressed them: relative to its scale, QT moves yields less than QE, because by the tightening episodes the float had shifted toward the elastic sectors, mutual funds and the arbitrageurs, so each unit of supply met a flatter demand curve. This is the composition-based asymmetry that Eren et al. (2026) emphasize; the yield level, by contrast, moves the measured elasticity but cancels in market clearing, so it shapes the proportional repricing, not the basis-point impact (Section 4). Looking forward, I fix the 2025Q1 portfolio and simulate further runoff at the Fed's 2024 pace. A further \$1 trillion would raise 10-year Treasury yields by 24 basis points and agency MBS yields by 18, and an aggressive \$1.6 trillion by 39 and 26. These are economically meaningful magnitudes: applied to \$27 trillion of marketable federal debt as it rolls over, the Treasury rise adds roughly \$65 billion a year to federal interest costs, and since agency MBS yields pass through nearly one for one to primary mortgage rates, the MBS rise adds about \$47 a month to the payment on a newly originated \$400,000 thirty-year fixed-rate mortgage. The marginal Treasury impact is nearly constant, about 2.4 basis points per \$100 billion, because Treasury issuance barely responds to yields.

This paper contributes to the growing literature on demand-based asset pricing in bond markets, with implications for unconventional monetary policy.[2] Koijen et al. (2021) find that foreign investors, whose demand is most elastic, sell the largest share of bonds to the ECB during quantitative easing. Jansen et al. (2024) estimate sector-level Treasury demand within an equilibrium model of arbitrageurs and find that QE moves yields only insofar as investors perceive the purchases as persistent; my counterfactuals, which hold the balance-sheet change in place, correspond to their credible-commitment case. Eren et al. (2026) estimate demand for U.S. Treasuries, find that QE and QT have asymmetric yield effects, and extend

[2] Several papers estimate demand systems for bond markets without a direct focus on QE: Fang et al. (2025) for global sovereign debt and Bretscher et al. (2025) for corporate bonds.

the analysis to Japan and the U.K. Fang and Xiao (2025) extend the approach to three bond classes at the security level and find that for every unit of bonds the Fed purchases, mutual funds and banks purchase an additional 0.5 units, amplifying the effect of Fed policy; their granularity also recovers substitution my sector-level classes average over, such as between long-maturity Treasuries and agency MBS, close substitutes at matched duration. Chaudhary et al. (2026) decompose realized Treasury yield movements into investor-level drivers with a granular instrumental-variable estimator and find that aggregate Treasury demand is highly inelastic: a 1 percent shift in demand moves yields by 13 basis points, consistent with the elasticities estimated here. My contribution is threefold. First, I let demand adjust at a sector-specific speed, estimating the horizon-by-horizon response by local projections, so the adjustment paths are estimated rather than imposed through an inertia parameter as in van der Beck (2026). The estimated dynamics separate the impact and long-run effects of QE and yield a new result: a balance-sheet shock's initial yield impact overshoots its long-run effect and dissipates as slow capital arrives (Duffie, 2010). Second, I measure each sector's demand against its own outside asset, where prior demand systems confine substitution to the bond universe, and identify the system with a granular instrument built from the same holdings data, which extends the sample to 1985 and brings QE1 in-scope. Third, I quantify the cross-asset spillovers of QE across four bond classes, including the first demand-system estimates for municipal bonds, an asset class absent from prior work, and find that the transmission is narrow: the spillovers are real but an order of magnitude below the own-market effects, evidence for the narrow view of Krishnamurthy et al. (2013) over a broad duration channel and a guide to which market the Fed must buy to move a given yield.

This paper also contributes to the empirical literature evaluating the effects of QE on asset prices. Event studies find that announcements of the Fed's purchase programs lowered 10-year Treasury yields by roughly 90 to 110 basis points around QE1 and mortgage rates by 40 to 100 basis points (Fuster and Willen, 2010; Gagnon et al., 2011; Hancock and

Passmore, 2011; Krishnamurthy and Vissing-Jorgensen, 2011; Stroebel and Taylor, 2012; Krishnamurthy et al., 2013), although Christensen and Rudebusch (2012) attribute about 60 percent of the QE1 decline to lower expected short rates, so announcement effects mix the signaling and portfolio-balance channels. Time-series and term-structure studies map the Fed's holdings into term premiums and find declines of roughly 30 to 45 basis points per program (D'Amico et al., 2012; Ihrig et al., 2018; Wei, 2022), and evidence from actual purchases, rather than announcements, points to persistent stock effects (D'Amico and King, 2013; Vissing-Jorgensen, 2021). These estimates, however, evaluate only the policies actually implemented, under the conditions prevailing at the time: they cannot speak to counterfactual policies, such as alternative purchase sizes or asset mixes, and are vulnerable to the Lucas critique, since reduced-form price responses need not be stable across policy regimes. I fill this gap by estimating structural demand parameters that are more stable across policy regimes and support counterfactual analysis for policies never implemented. Calibrated preferred-habitat models also support counterfactuals (Vayanos and Vila, 2021); relative to their calibration, which matches yield moments in a Treasury-only model, I estimate demand from observed sector portfolios and cover four asset classes, so the counterfactuals inherit measured investor heterogeneity and cross-asset substitution.

This paper also contributes to the literature on the transmission channels of balance sheet policy. QE transmits through signaling, portfolio rebalancing, and asset-specific channels such as safety, prepayment risk, and default risk (Krishnamurthy and Vissing-Jorgensen, 2011), as well as through bank lending (Chakraborty et al., 2020; Kandrac and Schlusche, 2021) and mortgage refinancing (Di Maggio et al., 2020). This paper focuses on portfolio rebalancing: when the Fed purchases one asset class, investors shift into imperfect substitutes, moving the yields of assets the Fed does not purchase. Selgrad (2023) finds direct evidence that Fed purchases of Treasuries and MBS induce investors to rebalance into corporate bonds. I quantify the channel by estimating substitution elasticities across four bond classes in a demand system. The estimates speak to the debate between a broad duration channel, under which purchases in any market move all long-term yields (Gagnon et al., 2011), and the narrow view that the impact concentrates in the purchased asset (Krishnamurthy et al., 2013): the rebalancing spillover is real, and larger when the Fed buys multiple asset classes at once, but it is an order of magnitude smaller than the own-market effect, so the transmission is mostly narrow.

Finally, this paper contributes to the literature on the implementation and limits of QT. How far the central bank can shrink its balance sheet without side effects depends on both sides of that balance sheet. On the liability side, QT drains reserves from financial intermediaries and strains the repo market, so the demand for reserves and liquidity bounds the size of QT (e.g., López-Salido and Vissing-Jorgensen, 2025; Anbil et al., 2026; Anderson et al., 2026; Oguri and Pizzimenti, 2026). This paper focuses instead on the asset side: as the Fed reduces its holdings of Treasuries and agency MBS, the remaining investors absorb the supply only at higher yields, and this yield effect, not reserve scarcity, limits QT. The asymmetry between QE and QT arises on both sides of the balance sheet (Acharya et al., 2024; Eren et al., 2026), so the effects of QT cannot be inferred from estimates of QE. Assessing QT therefore requires knowing who holds bonds, how steeply each holder's demand slopes, and how fast their capital moves; the demand system estimated here provides exactly these inputs.

# 2 Data

In this section, I outline the dataset. The dataset covers four key information required in the standard demand estimation technique used in the literature: portfolio share, asset prices (yields), asset characteristics, and instrumental variables for asset price.

## 2.1 Portfolio Weight

I use the Federal Reserve's Flow of Funds (the Financial Accounts of the United States) to compute sector-level portfolio shares. The Flow of Funds reports quarterly, market-value balance sheets for the U.S. financial sectors (e.g., life insurance companies, private pension funds). The sample runs from 1985Q1 to 2025Q1, a span that reaches back to the high-yield years of the late 1980s, which, as Section 3.6 shows, are what identify the yield response of the slow long-term investors. Over this span I work with 26 sectors. For the post-2010 period I add two refinements, where the data support them: I split foreign holdings into Foreign Official and Foreign Private using the Federal Reserve's CSLT estimates, and I add hedge funds, which the Flow of Funds does not report separately, from the Federal Reserve's Enhanced Financial Accounts (EFA), built from SEC Form PF filings. Neither is available before 2010, so the extended sample combines foreign holdings into one sector and keeps hedge funds within the household residual; I use the finer 28-sector split as a post-2010 robustness check. Table 1 lists the sectors and their broader sector-type classifications (e.g., insurers, pension funds).

Four major U.S. bonds (Treasuries, agency MBS, corporate bonds, and municipal bonds) are the assets inside the demand system; the Flow-of-Funds asset classes are coarser than CUSIP-level data but detailed enough (e.g., Treasury securities, corporate equity, cash) to identify them. Together the sectors form a heterogeneous demand side, yet a few large groups hold most of the bonds: throughout the sample, foreign investors and domestic financial institutions (banks, insurers, pension funds, and mutual funds) hold about 80% of the four bond markets, while arbitrageurs (e.g., dealers, hedge funds) hold only about 5%. Figure 1 reports each market's ownership composition and total issuance over time, from 1985 to 2025.

The outside asset is sector-specific. For each sector I designate a single non-bond asset class as the outside asset, the main alternative it trades off against bonds, and I measure its bond demand relative to that class, which forms the denominator of the sector's portfolio

Table 1: **Sector Coverage and Outside Asset Definitions**

| Type | Sector | Outside Asset |
|---|---|---|
| Banks | U.S.-Chartered Depositories | Loans |
| | Foreign Banking Offices | Total non-bond |
| | Credit Unions | Loans |
| | Banks in Affiliated Areas | Total non-bond |
| Insurers | Life Insurers | Corporate equities |
| | Property-Casualty Insurers | Corporate equities |
| Pension Funds | Private Pension Funds | Corporate equities |
| | State & Local Govt. Retirement Funds | Corporate equities |
| | Federal Govt. Retirement Funds | Total non-bond |
| Mutual Funds | Mutual Funds | Corporate equities |
| | Exchange-Traded Funds | Corporate equities |
| | Money Market Funds | Repo |
| | Closed-End Funds | Corporate equities |
| Arbitrageurs | Brokers and Dealers | Repo |
| | Hedge Funds | Repo |
| | mREITs | Total non-bond |
| Foreign | Foreign | Non-bond U.S. securities |
| Household Residual | Household Residual & Nonprofits | Corporate equities |
| Other | Nonfinancial Business | Total non-bond |
| | Government-Sponsored Enterprises | Total non-bond |
| | Holding Companies | Total non-bond |
| | State & Local Governments | Total non-bond |
| | Federal Government | Total non-bond |
| | Agency/GSE Mortgage Pools | Total non-bond |
| | Finance Companies | Total non-bond |
| | ABS Issuers | Total non-bond |
| | Other Financial Business | Total non-bond |

*Notes:* This table lists the sectors in the portfolio data. The outside asset is the holder-specific alternative against which bond demand is measured (the demand-system denominator): corporate equities for long-term investors, repo for money funds, dealers, and hedge funds, loans for depositories, and non-bond U.S. securities for foreign holders; "Total non-bond" denotes the residual of total financial assets net of the four bond classes. Hedge funds—which dominate the household Treasury residual through the cash-futures basis trade—are separated out using the Enhanced Financial Accounts (EFA) and modeled in Treasuries only. Institution-type classifications are defined by the author; Appendix D.1 explains the outside-asset choice for each sector. Data sources and Flow-of-Funds table codes for each sector are listed in Appendix Table A.9.

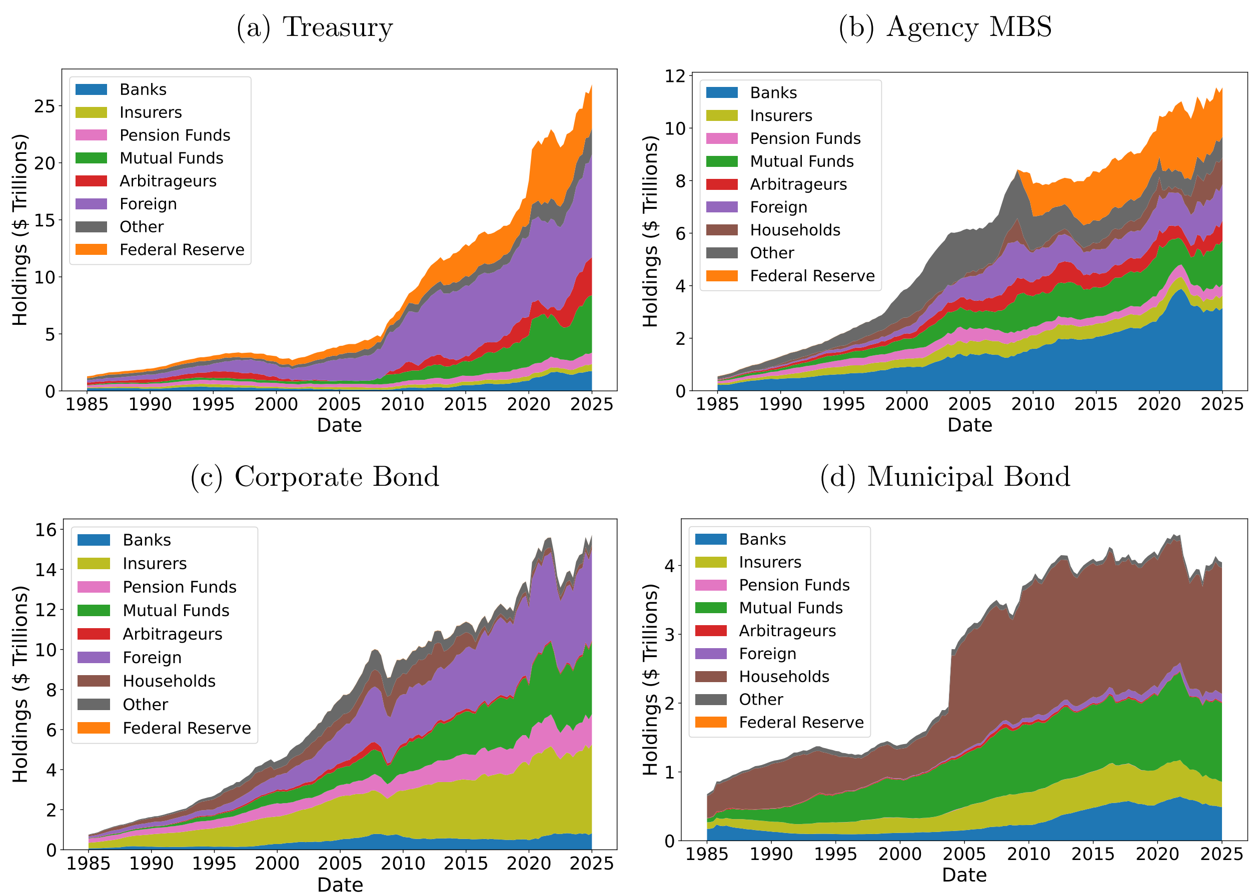
share in the demand system. A single class captures the margin along which a sector actually rebalances, whereas the residual of all non-bond holdings would lump in assets it does not trade off against bonds. Domestic banks, for example, rebalance between bonds and loans, not between bonds and every non-bond asset on their balance sheet, so I use loans as their outside asset. I describe the outside-asset choice for each sector in Section 3, and Appendix D.1 records the rationale sector by sector; the last column of Table 1 reports each sector's outside asset. Figure A.6 in the Appendix shows how each sector's portfolio shares evolve over time.

I acknowledge the limitations of the sector-level portfolio share data from the Flow of Funds. The data do not report within-class maturity variation, so I cannot estimate demand across precise duration buckets. Thus, this paper focuses on substitution across broad fixed-income asset classes rather than fine maturity choice.

## 2.2 Asset Characteristics

**Yields.** This paper uses one yield per asset class as its price. For Treasuries and corporate bonds I use representative constant-quality series that span the full sample: the 10-year constant-maturity Treasury yield (DGS10) and Moody's seasoned Baa corporate yield, both from FRED. A point yield at a representative maturity tracks the long end of each market, where habitat demand bites, more cleanly than a whole-market index does over a four-decade sample. For agency MBS I use an option-adjusted yield, the 10-year Treasury yield plus the option-adjusted spread of the Bloomberg agency-MBS index (available from 1990Q3, which sets the start of the MBS estimation sample): the index's own yield-to-worst embeds the prepayment-option premium, which moves with interest-rate volatility rather than with the compensation investors require for holding MBS, and stripping it sharpens the identification (the first-stage $F$ of the MBS supply instruments rises from about 6 to 11). For municipal bonds I use the yield of the Bloomberg municipal index, the only continuous series over the sample. Table A.10 in the Appendix reports the sources and ticker codes. I aggregate all

Figure 1: **Ownership Structure of U.S. Bond Markets**




*Note:* This figure shows the dynamics of the ownership share structure of the U.S. fixed income market by institutional sector, covering Treasury, agency MBS, corporate bond, and municipal bond. The classification of institutional sectors is described in Appendix A.1. In Panel (a), the Z.1 household residual in Treasury securities is reassigned to hedge funds within the Arbitrageurs sector, as these holdings predominantly reflect hedge funds engaged in the cash-futures basis trade. In Panel (d), the sharp increase in municipal securities outstanding between 2003:Q4 and 2004:Q1 reflects a statistical revision in the Z.1 dataset rather than an actual surge in issuance.

series to the quarterly frequency, taking the end-of-quarter value.

**Other Characteristics.** This paper includes standard fixed-income asset characteristics: maturity, issuance size, duration, and coupon. I obtain the weighted time-to-maturity, modified duration, and weighted-average coupon of each index from Bloomberg. Modified duration measures the percentage price change of the index for a one-percentage-point change in yield, and thus captures the interest rate sensitivity of each asset class; because the change in modified duration is a near-mechanical function of the contemporaneous yield change ($dD \approx -D\,dy/(1+y)$; the correlation of the two quarterly changes is $-0.9$ for Treasuries), duration enters the estimation lagged one quarter, as a predetermined characteristic. The coupon, a slow-moving composition variable, serves as an additional exogenous shifter in the construction of the heteroskedasticity instruments (Section 4). I also obtain the issuance size of each asset class from the Z.1 Financial Accounts dataset and enter it as a log-transformed control; the Financial Accounts report outstanding amounts at market value, so the differenced control captures net issuance together with revaluation, a reduced-form quantity measure. The municipal index maturity is missing (reported as zero) for 1986Q1–1987Q4; I interpolate linearly across the gap.

For the post-2010 robustness specification (Appendix D.4) I also construct a liquidity measure per asset class, the proportional bid-ask spread for Treasuries from CRSP and the Jankowitsch et al. (2011) price dispersion from TRACE and MSRB trade data for the other assets, standardized within asset class; the main specification does not use it. Table A.10 in the Appendix summarizes the data sources for each asset class.

**Macroeconomic Variables.** I include a set of macroeconomic controls to absorb time-series variation in aggregate portfolio demand unrelated to asset prices. Specifically, I control for log real GDP, real GDP growth, core inflation, and the log of the U.S. dollar index, all obtained from the Federal Reserve Economic Data (FRED) database and aggregated to the quarterly frequency. I also include the one-quarter lag of log VIX as a measure of aggregate risk sentiment. Because VIX is jointly determined with asset prices during flight-to-safety episodes, I use the lagged value as a predetermined control to avoid simultaneity bias.

## 2.3 Instruments

The main instruments are a granular instrument built from the Flow-of-Funds holdings for Treasury yields, the Fed's SOMA purchases and the Drukker (2025) surprise for agency MBS yields, and heteroskedasticity-based instruments for corporate and municipal yields, as explained in Section 3; maturity walls and the monetary-policy surprises of the prior demand-system literature enter as robustness checks. This subsection records the data sources.

**Granular Instrument.** I build the granular instrument of Section 3 from the same sector-level Flow-of-Funds holdings used for the portfolio shares, so it needs no data beyond the Financial Accounts. I extract every sector's quarterly idiosyncratic Treasury-holding-growth shock, the residual from a common-factor projection, and aggregate the size-weighted shocks across sectors, excluding the sector being instrumented. The Lewbel instruments for corporate and municipal yields are likewise internal, built from the asset characteristics already in the panel.

**Monetary Policy Announcement Shocks.** For the post-2010 robustness specification I use the high-frequency monetary policy shocks of Swanson (2021) and Drukker (2025), identified from intraday financial-market movements around Federal Open Market Committee (FOMC) announcements. These shocks isolate the unexpected component of monetary policy, which instruments Treasury and agency MBS yields.

**Corporate and Municipal Bond Maturities.** I construct maturity-wall instruments for corporate and municipal bonds using CUSIP-level bond maturity schedules. Corporate bonds are drawn from Mergent FISD, excluding convertible, putable, and asset-backed securities, and span 2000Q1–2024Q4, averaging \$212 billion per quarter. Municipal bonds are drawn from the MSRB transactions database, where offering size is approximated by par traded within the first 30 days of issuance, and span 2009Q1–2024Q4, averaging \$9.4 billion per quarter.[3]

**Federal Reserve Asset Purchases.** I also use detailed records of Fed asset purchases provided by the Federal Reserve Bank of New York. The dataset tracks the Fed's purchases and sales of Treasury and agency MBS securities over time, including the timing and scale of each QE round: QE1 (2008Q4–2010Q1), QE2 (2010Q4–2011Q2), QE3 (2012Q3–2014Q4), and QE4 (2020Q1–2021Q4). I use these records in two ways: as a supply-side instrument in the post-2010 robustness specification, and, centrally, to define the balance-sheet shocks in the counterfactual of Section 5.

## 2.4 Summary Statistics

Table 2 reports summary statistics for the merged dataset. Panel (a) presents average portfolio shares and assets under management (AUM) by institution type. Foreign investors allocate the largest share of their portfolios to Treasuries (20.7%), while insurers concentrate heavily in corporate bonds (32.6%). Households and pension funds hold relatively small fixed-income shares despite their large aggregate AUM. Ownership is concentrated in weakly yield-sensitive hands: banks, insurers, pension funds, foreign investors, and the household residual together hold about four-fifths of the four bond markets, while the sectors that turn out to be elastic, mutual funds and the arbitrageurs, hold about a fifth. This concentration is what makes the aggregate demand curve steep: the elastic capital is too small a share to lift the aggregate, whose elasticity is governed by the large, yield-insensitive holdings. Panel (b) reports asset characteristics across the four bond classes over the sample period. Treasury and agency MBS yields average 4.7%, corporate bonds the highest at 7.0%, while municipal bonds offer the lowest average yield (4.3%), reflecting their tax-exempt status. Corporate

[3]The municipal instrument begins in 2009Q1 because bonds maturing earlier were issued before MSRB coverage began in 2005, precluding recovery of their offering sizes.

bonds have the longest average maturity (11.5 years) and agency MBS the shortest duration (4.8 years), consistent with prepayment risk compressing effective duration. The estimation sample covers 26 sectors observed quarterly from 1985Q1 to 2025Q1.

# 3 Empirical Framework

This section develops a demand system for four U.S. bond assets (Treasuries, agency MBS, corporate bonds, and municipal bonds) identified from sector-level portfolio-share data. Two features set it apart from existing bond demand systems: the outside asset differs across sectors, and demand adjusts at a sector-specific speed. Existing systems typically measure substitution within the bond universe or against a single, common outside asset; I instead let the outside asset differ across sectors, measuring each sector's bond demand against the non-bond class it actually rebalances toward, for example, loans for banks, corporate equities for long-term investors, and repo for money funds and dealers. The bank choice, for instance, is supported by evidence that banks rebalance between loans and bonds under QE (e.g., Rodnyansky and Darmouni, 2017; Kandrac and Schlusche, 2021; Kurtzman et al., 2022). Table 1 lists the outside asset for each sector.

This sector-specific choice of outside assets has a cost and a benefit. The cost is that the estimated demand parameters (the yield and characteristic coefficients) are no longer directly comparable across sectors, since each is defined relative to a different outside asset. The benefit is that the implied demand elasticity remains comparable across sectors and aggregates cleanly; the elasticity, not the raw coefficients, is what the counterfactual requires. The design also widens the substitution margin: because the outside assets span loans, equities, and repo, the system measures substitution across the full asset spectrum, not only within bonds. It therefore captures the portfolio-balance channel of QE and QT, through which central-bank purchases and sales reallocate investors between bonds and other assets. Jansen et al. (2024), Fang et al. (2025), and Eren et al. (2026) measure demand against other

Table 2: **Summary Statistics**

(a): Portfolio Summary

| | Financial Assets ($T) | | Portfolio Weight (%) | | | |
|---|---|---|---|---|---|---|
| Sector Type | Total | Bond | Treasury | Agency MBS | Corporate | Municipal |
| Banks | 12.87 | 2.74 | 3.60 | 10.78 | 4.21 | 2.12 |
| Insurers | 6.24 | 2.75 | 4.81 | 7.08 | 33.12 | 1.04 |
| Pension Funds | 13.21 | 1.51 | 4.29 | 3.05 | 6.36 | 0.02 |
| Mutual Funds | 12.07 | 3.74 | 7.61 | 7.13 | 9.09 | 8.34 |
| Arbitrageurs | 3.22 | 0.56 | 3.46 | 11.50 | 3.57 | 0.00 |
| Foreign | 19.68 | 10.96 | 32.60 | 4.85 | 19.75 | 0.00 |
| Household Residual | 50.04 | 2.50 | 1.09 | 0.49 | 1.32 | 2.72 |
| Other | 30.91 | 2.01 | 2.23 | 3.72 | 1.49 | 0.23 |

(b): Asset Summary

| Asset | Variable | Mean | Median | S.D. | Min | Max | N |
|---|---|---|---|---|---|---|---|
| Treasury | Yield (%) | 4.77 | 4.44 | 2.39 | 0.66 | 11.65 | 161 |
| | Maturity (years) | 8.06 | 8.16 | 0.94 | 6.08 | 9.88 | 161 |
| | Issuance (log $M) | 15.50 | 15.16 | 0.89 | 14.06 | 17.11 | 161 |
| | Duration (years) | 5.49 | 5.35 | 0.63 | 3.93 | 7.11 | 161 |
| | Coupon (%) | 5.29 | 4.97 | 2.83 | 1.57 | 11.55 | 161 |
| Agency MBS | Yield (%) | 4.71 | 4.54 | 2.03 | 1.31 | 9.60 | 139 |
| | Maturity (years) | 6.75 | 6.78 | 1.63 | 2.15 | 12.31 | 161 |
| | Issuance (log $M) | 15.28 | 15.63 | 0.86 | 13.22 | 16.26 | 161 |
| | Duration (years) | 4.76 | 4.84 | 0.84 | 1.84 | 6.38 | 161 |
| | Coupon (%) | 6.06 | 5.50 | 2.32 | 2.60 | 10.73 | 161 |
| Corporate | Yield (%) | 7.07 | 6.71 | 2.19 | 3.11 | 13.69 | 161 |
| | Maturity (years) | 11.52 | 11.03 | 1.41 | 9.42 | 15.04 | 161 |
| | Issuance (log $M) | 15.54 | 15.86 | 0.86 | 13.55 | 16.58 | 161 |
| | Duration (years) | 6.60 | 6.36 | 0.74 | 5.56 | 8.75 | 161 |
| | Coupon (%) | 6.40 | 6.15 | 1.91 | 3.55 | 9.66 | 161 |
| Municipal | Yield (%) | 4.33 | 4.08 | 1.87 | 1.00 | 9.64 | 161 |
| | Maturity (years) | 14.09 | 13.51 | 1.69 | 12.77 | 20.33 | 161 |
| | Issuance (log $M) | 14.63 | 14.91 | 0.58 | 13.43 | 15.31 | 161 |
| | Duration (years) | 6.07 | 5.99 | 0.92 | 4.73 | 8.06 | 161 |
| | Coupon (%) | 5.66 | 5.13 | 1.23 | 4.45 | 9.03 | 161 |

*Notes:* Panel (a) reports sector-level financial assets and portfolio allocations across four fixed-income asset classes, drawn from the Z.1 Flow of Funds. Portfolio weights are time-series averages, weighted by sector financial assets. Panel (b) reports summary statistics for yield, maturity, issuance, duration, and coupon at the asset–quarter level, the unit at which the characteristics vary ($N$ is the number of quarters; the agency-MBS yield begins in 1990Q3).

bonds, substitution within fixed income; my framework measures it against non-bond assets, the margin the portfolio-balance channel operates on. This changes the elasticity, not just the margin: investors substitute far more readily between one bond and another than between bonds and equities or loans, so a within-bond outside asset yields a more elastic demand than the bond-versus-non-bond margin that governs the aggregate price impact of central-bank purchases. The assignment shapes who responds, not how much yields move: replacing every sector's outside asset with the generic non-bond residual cuts the measured insurer slope by nearly half yet moves the baseline QT counterfactual by less than two basis points, because the sector-level shifts offset in the holdings-weighted aggregate (Appendix D.1).

The demand system so far is static, describing the allocation each sector holds once it has fully adjusted. Investors do not rebalance instantly, however, so I also estimate how fast each sector moves, tracing the horizon-by-horizon response to a yield change by local projections. This separates the impact slope from the long-run slope: the long-run slope is the demand parameter the counterfactual uses, while the estimated path governs the transition to the new equilibrium. The largest long-term investors turn out to be the slowest, a dimension of heterogeneity as consequential for QE and QT as the level of the elasticity itself.

Technically, the use of aggregate shares is analogous to the market-level setup of the standard BLP framework (Berry et al., 1995). I specify a multinomial logit demand system following the asset-demand literature, allowing for heterogeneity across sectors. Unlike standard BLP applications, within-sector heterogeneity is unidentifiable in the asset-demand context, since all sectors face the same asset prices and characteristics at each point in time. The model therefore reduces to a special case of BLP without random coefficients, as described in Appendix A.

## 3.1 Setup

I index sector-level investors by $i = 1, \ldots, I$; sectors include, for example, life insurers and mutual funds. Assets are indexed by $n = 0, \ldots, N$, where $n = 0$ denotes the outside asset, which differs across sectors and comprises non-bond holdings such as equities and loans. The bond universe consists of Treasury securities, agency MBS, corporate bonds, and municipal bonds. The yield of asset $n$ in quarter $t$ is $y_t(n)$. Each asset has a vector of observed characteristics $x_t(n)$: maturity, issuance, and duration. The characteristics enter the estimation predetermined; in particular, duration enters lagged one quarter, because the change in modified duration is a near-mechanical function of the contemporaneous yield change (Section 2). The portfolio weight of investor $i$ in asset $n$ at time $t$ is $w_{it}(n)$, with $\sum_{n=0}^{N} w_{it}(n) = 1$.

## 3.2 Portfolio Choice

I assume that portfolio shares follow a characteristics-based logit demand, under which the share is exponential in a linear index of characteristics (Koijen and Yogo, 2019).[4] The portfolio weight $w_{it}(n)$ is

$$w_{it}(n) = \frac{\delta_{it}(n)}{1 + \sum_{n=1}^{N} \delta_{it}(n)}, \tag{1}$$

where

$$\delta_{it}(n) = \exp\Big\{\beta_i y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n)\Big\}\epsilon_{it}(n). \tag{2}$$

$\beta_i$ denotes investor $i$'s demand semi-elasticity with respect to the yield, and $\gamma_i$ (a vector) collects investor $i$'s loadings on asset characteristics. $\xi_{it}(n)$ denotes an unobserved demand shifter capturing asset-time variation not explained by observed characteristics. $\epsilon_{it}(n)$ is the latent demand not explained by the observed yield or characteristics. Note that the outside-asset portfolio weight is given by

$$w_{it}(0) = \frac{1}{1 + \sum_{n=1}^{N} \delta_{it}(n)}. \tag{3}$$

[4] Koijen and Yogo (2019) provide an asset pricing–based theoretical foundation for logit demand by deriving it from mean–variance portfolio choice under factor-structure assumptions.

Combining (1) and (3) yields the following estimable specification:

$$\begin{aligned}\ln \frac{w_{it}(n)}{w_{it}(0)} &= \ln \delta_{it}(n) \\ &= \beta_i y_t(n) + \gamma_i' x_t(n) + \xi_{it}(n) + \varepsilon_{it}(n).\end{aligned} \tag{4}$$

The left-hand side is the log portfolio share of asset $n$ relative to the outside asset, which is observed in the data. The right-hand side is a linear combination of the asset's yield $y_t(n)$, characteristics $x_t(n)$, and asset–time demand shifter $\xi_t(n)$ with estimable coefficients. $\varepsilon_{it}(n)$ is an idiosyncratic demand shock, which is log of the latent demand $\epsilon_{it}(n)$. I parameterize the asset–time demand shifter $\xi_t(n)$ as

$$\xi_{it}(n) = \alpha_n + \nu_{in}' W_t + \tilde{\xi}_{it}(n), \tag{5}$$

where $W_t$ is a vector of macroeconomic controls, $\nu_{in}$ captures sector- and asset-specific loadings on macro conditions, and $\tilde{\xi}_{it}(n)$ is the residual unobserved demand shifter. I follow Jansen et al. (2024) and Eren et al. (2026) in parameterizing the asset–time demand shifter via macro controls rather than asset–time fixed effects, which would absorb the time-series variation in the instruments.

## 3.3 Rebalancing Speed

The estimation is in first differences. Differencing the logit demand (4) is a pure algebraic transformation that removes the asset fixed effect $\alpha_n$ while leaving every slope unchanged,

$$\Delta \ln \frac{w_{it}(n)}{w_{it}(0)} = \beta_i \, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \Delta \varepsilon_{it}(n), \tag{6}$$

and, because it regresses changes on changes, avoids the spurious-regression problem that the trending yield level poses over a four-decade sample.

Equation (6) recovers the semi-elasticity $\beta_i$, however, only if sectors complete their rebalancing within the quarter, and the speed of rebalancing differs sharply across sectors. Insurers and pension funds complete only a fraction of their rebalancing within the quarter, held back by asset–liability mandates, illiquid positions, and the slow-moving capital of Duffie (2010); banks reach their long-run slope within a quarter or two, and even the arbitrageurs, the fastest movers, complete only part of their response on impact. Estimated on one-quarter changes, (6) therefore delivers only the *impact* response, near zero for a slow sector even when its full response is large. The framework already gives each sector its own outside asset, the margin along which it rebalances; I now give each sector its own speed.

Sluggish adjustment is not an implication of the static logit, which describes the portfolio a sector ultimately wants; it is an assumption about how observed holdings reach that target. I treat Equation (4) as the target portfolio and let observed shares converge to it over time, so the horizon-$h$ slope measures how much of the move toward the new target is complete $h$ quarters after the shock. This is the structure van der Beck (2026) formalizes for equity institutions, where demand elasticities rise with the horizon as slow capital arrives, and its limit is the static parameter: under instantaneous adjustment the horizon slopes are flat at $\beta_i$. Appendix D.6 makes the bridge explicit, showing that a geometric partial-adjustment model toward the logit target nests this design and matches the estimated insurer path, while the local projections below recover the adjustment path without imposing its shape.

To do so, I measure, for each horizon $h = 0, 1, 2, \ldots$, how far the portfolio has moved $h$ quarters *after* the yield change, with instrumental-variables local projections (Jordà, 2005):

$$\sum_{s=t}^{t+h} \Delta \ln \frac{w_{is}(n)}{w_{is}(0)} = \beta_i^{(h)}\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \tau_{in}\, t + u_{it}^{(h)}(n), \tag{7}$$

where $\Delta y_t(n)$ is instrumented by the asset's instrument set (Section 3.6), the linear trend $\tau_{in}\, t$ absorbs the secular comovement of the long sample, and the moving-average error $u^{(h)}$ calls for Newey–West standard errors, with the bandwidth set to the horizon $h$ (Newey and West, 1987). The left-hand side telescopes to the change in the log share between quarters

$t-1$ and $t+h$, so (7) regresses an $(h+1)$-quarter change of the share on a one-quarter change of the yield; the differencing window of the yield stays fixed, only the response window of the share extends. This is deliberately *not* a long-difference specification, which would also difference the yield over $h+1$ quarters: lengthening the yield window lets the four-decade secular comovement of yields and portfolio shares back into the regressor, the spurious-trend problem that differencing was meant to remove. At $h=0$, (7) collapses back to (6) and $\beta_i^{(0)}$ is the impact slope; as $h$ grows, $\beta_i^{(h)}$ rises toward the sector's full response and flattens once rebalancing is complete.

The long-run slope, which I read one year out at $h=4$ once the response has flattened, is the semi-elasticity $\beta_i$ of the logit demand (4). The identification requires no particular adjustment process: whatever the frictions, once rebalancing has run its course the allocation is the one the static system describes, so the cumulative response flattens exactly at the static slope. The speed of adjustment is a rebalancing friction, not a demand parameter: the preferences in (4) and every object built on them, the elasticities, market clearing, and the counterfactual equilibrium, are unchanged, and the frictions govern only how quickly observed holdings reach the allocation that (4) prescribes. A slow sector barely moves on impact even when its long-run slope is large, so a demand curve estimated from quarterly changes alone understates how far these investors ultimately rebalance.[5] The counterfactual's long-run equilibrium uses $\beta_i$; the transition to it uses the estimated path $\beta_i^{(h)}$. Insurers and pension funds, flat on impact, recover positive long-run slopes within a few quarters, while banks adjust faster (Section 4).

In all demand estimation, the share changes enter valuation-adjusted: because the Financial Accounts report holdings at market value, I follow Eren et al. (2026) and replace $\Delta \ln\{w_{it}(n)/w_{it}(0)\}$ with the adjusted change $\Delta q_{it}(n)$, which strips the mechanical mark-to-

[5] This reconciles why levels and difference estimates of the same demand equation disagree. A levels regression weights low-frequency variation, where adjustment is complete, and recovers a slope near the long-run $\beta_i$, but conflates it with the four-decade secular comovement of yields and shares and needs a near-unit-root instrument whose relevance can be spurious; a quarterly difference weights high-frequency variation and recovers the impact slope $\beta_i^{(0)}$. The local projections (7) recover the long-run slope from differenced, trend-controlled data, avoiding both, and unlike a single levels coefficient trace the adjustment path.

market component of the share change and retains active rebalancing (Appendix D.3).

## 3.4 Demand Elasticity

The logit demand system yields a closed-form expression for the yield elasticity of demand, the key object behind the aggregate demand curve. The elasticity of the portfolio share of asset $n$ with respect to the yield of asset $m$ is

$$\eta_{it}(m,n) = \frac{\partial \log w_{it}(n)}{\partial \log y_t(m)} = \begin{cases} \beta_i \cdot \big(1 - w_{it}(n)\big) \cdot y_t(n) & m = n \\ -\beta_i \cdot w_{it}(m) \cdot y_t(m) & m \neq n, \end{cases} \tag{8}$$

where $\beta_i$ is the long-run yield sensitivity of sector $i$ from Section 3.3, $w_{it}(n)$ is the portfolio share of asset $n$, and $y_t(n)$ is the yield of asset $n$ in quarter $t$. The own-elasticity ($m = n$) gives the percentage change in the portfolio share per one-percent rise in the asset's own yield; the cross-elasticity ($m \neq n$) gives the spillover to other assets.

A related object is the bond-demand elasticity, the percent change in total bond holdings per one-percent rise in the average yield:

$$\eta_{it}^{\text{bond}} = \frac{\partial \log w_{it}^{\text{bond}}}{\partial \log \bar{y}_t} = \beta_i \cdot \big(1 - w_{it}^{\text{bond}}\big) \cdot \bar{y}_t, \tag{9}$$

where $w_{it}^{\text{bond}} = \sum_{n=1}^{N} w_{it}(n)$ is the total bond share (the outside asset is $n = 0$) and $\bar{y}_t$ is the average yield across the four asset classes. Unlike Equation (8), the term $(1 - w_{it}^{\text{bond}})$ here equals the outside-asset share only, as within-bond substitution cancels out. The aggregate market elasticity is the bond-AUM-weighted average across sector types:

$$\eta_t^{\text{mkt}} = \sum_i \alpha_{it} \cdot \eta_{it}^{\text{bond}}, \tag{10}$$

where $\alpha_{it} = w_{it}^{\text{bond}}\,\text{AUM}_{it} / \sum_j w_{jt}^{\text{bond}}\,\text{AUM}_{jt}$ denotes the bond-AUM weight of sector type $i$.

## 3.5 Market Clearing

To close the asset pricing model, the market for each asset $n$ clears in each period $t$:

$$\sum_i A_{it} \cdot w_{it}(n) = S_t(n), \tag{11}$$

where $A_{it}$ is investor $i$'s total wealth at time $t$ and $S_t(n)$ denotes the outstanding supply of asset $n$ (equivalently, its total market value) at time $t$. Market clearing holds in this study because, by construction, the aggregate share data cover all investors and sectors.

Together with the demand slopes, the clearing condition implies a price impact. Linearizing Equation (11) around the observed equilibrium, a supply shift $\Delta S_t(n)$ must be absorbed by the sectors' share adjustments, $\sum_i A_{it}\, \beta_i\, w_{it}(n)\big(1-w_{it}(n)\big)\, \Delta y_t(n) = \Delta S_t(n)$, so removing 1 percent of asset $n$'s private float raises its yield by

$$\Delta y_t(n) = \frac{1}{\sum_i s_{it}(n)\, \beta_i\, \big(1 - w_{it}(n)\big)} \text{ basis points}, \tag{12}$$

where $s_{it}(n) = A_{it}\, w_{it}(n)/S_t(n)$ is sector $i$'s share of the float. Unlike the elasticity in Equation (8), the yield level cancels: investors respond to yields point for point, so the basis-point impact is pinned down by this yield-free slope, and its movement over time reflects the composition of holdings alone. This expression is the first-order version of the counterfactual in Section 5, which clears the full nonlinear system.

## 3.6 Identification

The demand parameters are estimated from the first-difference specification (6) and, horizon by horizon, the local projections (7), which regress the change in the log portfolio share relative to the outside asset on the change in asset yields and characteristics. While prior studies often treat non-price characteristics as exogenous, the yield is likely endogenous because demand shocks can affect market prices. This endogeneity concern is amplified in this setting, as the model aggregates investors rather than assuming atomistic behavior, making simultaneity between demand shocks and yields non-negligible.

I identify the demand slope with external supply shocks where they exist and heteroskedasticity-based instruments where they do not. Treasury yields are instrumented by a granular instrument built from the sectors' own idiosyncratic rebalancing; agency MBS yields by the Fed's SOMA purchases together with the MBS-price surprise of Drukker (2025); and corporate and municipal yields, for which no relevant external supply shifter exists, by the heteroskedasticity-based instruments of Lewbel (2012). The exclusion restriction for the external instruments is that their variation moves portfolio shares only through yields, after controlling for macroeconomic conditions. Because the estimating equations are the differenced specification (6) and the local projections (7), whose horizon-$h$ error cumulates the latent-demand revisions from $t$ through $t + h$, the condition applies to those revisions over the projection window:

$$\mathbb{E}\big[\Delta\varepsilon_{i,t+s}(n) \mid Z_t(n)\big] = 0, \qquad s = 0, 1, \ldots, h. \tag{13}$$

At $h = 0$ this is the standard contemporaneous exclusion restriction; at longer horizons it adds the lead-exogeneity condition of LP-IV (Stock and Watson, 2018): the instrument must be unpredictable not only for current latent demand but for its subsequent revisions over the year the response cumulates. The high-frequency monetary-policy surprises of Swanson (2021) are weak over the 1985–2025 sample, because quarterly aggregation dilutes the FOMC-window surprises; they enter only as a post-2010 robustness check.

**Granular Instrument for Treasury and Agency MBS Yields** The largest holders of Treasuries and agency MBS are big enough that their idiosyncratic rebalancing moves the market yield. I use this granularity to build an instrument in the spirit of Gabaix and Koijen (2024). The idea is simple: when a large sector rebalances for reasons of its own, unrelated to marketwide conditions, it shifts the supply that every other investor must absorb and moves the yield. The idiosyncratic shocks of many small investors would wash out in the aggregate, but a handful of large, granular sectors leave a footprint in it, and that footprint is the variation the instrument isolates, exogenous to the common demand shock that also drives yields. Let $g_{jt}(n)$ be the idiosyncratic component of sector $j$'s holding growth in asset $n$, the residual from projecting sector-level holding changes on a common factor:

$$\Delta \ln H_{jt}(n) = \lambda_j(n)\, f_t(n) + g_{jt}(n), \tag{14}$$

where $H_{jt}(n) = w_{jt}(n)\, A_{jt}$ is sector $j$'s dollar holdings of asset $n$, its portfolio share times its assets under management, $f_t(n)$ is the common factor of holding growth across sectors, extracted as the first principal component, and $\lambda_j(n)$ is sector $j$'s loading. The instrument is built from dollar holdings rather than the share ratio in Equation (4) because what moves the market yield is the supply a large sector's rebalancing releases or absorbs, a dollar flow. The common factor absorbs the marketwide demand shock that moves all sectors together; what remains, $g_{jt}(n)$, is each sector's own rebalancing. The granular instrument for asset $n$ is the size-weighted sum of these idiosyncratic shocks, net of their cross-sectional mean and excluding the sector being instrumented:

$$\Gamma_t^{-i}(n) = \sum_{j \neq i} s_{j,t-1}(n)\, g_{jt}(n) \;-\; \frac{1}{J} \sum_{j \neq i} g_{jt}(n), \tag{15}$$

where $s_{j,t-1}(n)$ is sector $j$'s lagged share of asset $n$. Large sectors' idiosyncratic demand shifts move the aggregate yield, delivering relevance, but are by construction uncorrelated with sector $i$'s own demand shock, delivering the exclusion restriction.

The main threat to the exclusion is correlated substitution: sectors that rebalance along the same margin may load on a common substitution shock that the leave-one-out sum retains. Two features limit this. First, the common-factor projection that defines $g_{jt}(n)$ already removes the comovement shared across sectors, including the common component of any substitution wave. Second, the demand system's own outside-asset structure diversifies the donors: the sectors that dominate the size weights rebalance along different margins, banks against loans, long-term investors against equities, money funds against repo, so their idiosyncratic shocks have little reason to comove with sector $i$'s.[6]

Foreign Treasury demand corroborates the granular instrument's exogeneity. The rest of the world is the largest Treasury holder, and its demand is reserve-driven, accumulation for reasons orthogonal to U.S. domestic demand shocks, which makes the foreign idiosyncratic shock the most defensibly exogenous single donor. Estimating the Treasury slope with the foreign shock alone reproduces the granular-instrument estimate, and using both jointly passes the Sargan overidentification test (Section 4). Foreign demand for the *other* assets does not qualify: foreign MBS holdings are large but yield-chasing, bought when the spread over Treasuries is attractive, so they fail exogeneity, and foreign investors hold no tax-exempt municipal bonds.

**SOMA Purchases for Agency MBS Yields** The granular instrument is weak for agency MBS, where the large holders' flows track the mortgage cycle. Instead I use the Fed's own SOMA purchases of agency MBS, the quarterly change in Fed MBS holdings, as the supply shock: purchases are set by monetary-policy objectives, not by any private sector's MBS demand, and they remove supply from the private market directly. I supplement them with the MBS-price surprise of Drukker (2025), constructed from intraday Treasury futures and agency MBS transactions in 30-minute FOMC windows. SOMA purchases fail for Treasuries, where the Fed's purchase timing is correlated with the yield cycle, but for MBS the first stage is strong with the expected sign.

**Heteroskedasticity-Based Instruments for Corporate and Municipal Yields** No relevant external supply shifter exists for corporate and municipal yields. The granular in-

[6] A stricter construction that also excludes every sector sharing $i$'s outside-asset group is not viable as the baseline: the residual outside-asset group spans half the market, so for those sectors the exclusion removes most of the size weight and the first stage collapses. The reserve-driven foreign shock below provides the sharper external check instead.

strument has no bite in these markets, because no sector combines the size, the idiosyncratic flow volatility, and the exogeneity it requires: insurers hold the largest corporate stock, but their premium-driven flows are smooth, and what variation they have tracks the credit cycle; mutual-fund flows are volatile but return-chasing; and the remaining large holder is the household residual. Re-running the granular construction on these markets confirms a first stage near zero. Maturity walls, the face value of bonds scheduled to mature each quarter (Almeida et al., 2012; Harford et al., 2014), are predetermined but too small a share of yield variation to be relevant; realized gross issuance responds to rates; and foreign demand is either absent (municipals) or yield-chasing (corporates). I therefore identify these two slopes with the heteroskedasticity-based instruments of Lewbel (2012). At the asset–quarter level I regress the yield change on the differenced asset characteristics, take the residual $\hat{e}_t(n)$, and use the demeaned quantity-side characteristics, maturity, issuance, and coupon, interacted with that residual,

$$Z_t(n) = \left[\Delta x_t(n) - \overline{\Delta x(n)}\right] \hat{e}_t(n), \tag{16}$$

as instruments. Identification requires that the characteristics be uncorrelated with the product of the demand and yield shocks and that the yield-shock variance move with the characteristics, a condition bond markets satisfy through their volatility regimes. These are internal instruments, weaker in economic grounding than external supply shocks; I use them only for the two assets where no external shifter exists, and weak-instrument-robust inference applies throughout. The maturity walls remain as a robustness check.

Finally, the sample itself is part of the identification. The extended 1985–2025 span is not merely more data; it supplies the price variation that identifies the slow investors. The high-yield years of the late 1980s reveal the yield-seeking of insurers and pension funds, whose response is invisible at the near-zero yields that prevail after 2010. Estimating from 1990 instead collapses the insurer long-run slope toward its impact value; the slow-adjustment mechanism is identified by the 1980s.

# 4 Results

## 4.1 Identification Strength

Every asset's yield is identified. Table 3 reports the first stage on the 1985–2025 asset–quarter panel: the Treasury granular instrument has a first-stage $F$ of 12.1 with the expected sign, a buy shock lowers the yield ($t = -3.2$); the SOMA purchases carry agency MBS ($F = 11.0$, $t = -2.9$ on the option-adjusted yield); and the Lewbel (2012) instruments deliver $F$ of 7.3 for corporate and 7.6 for municipal yields, assets for which every candidate external shifter fails. Because the yield $y_t(n)$ varies only at the asset–quarter level, the first stage is common to every sector, so cross-sectional differences in the estimated slopes reflect genuine demand heterogeneity, not differences in the identifying variation. The joint $F$ across all nine instruments is 8.5; since the instruments are block-diagonal across assets, the per-asset statistics are the informative ones, and the system is as strong as its weakest block. The Treasury and agency MBS blocks clear the conventional threshold; the heteroskedasticity blocks approach it. First stages of this magnitude are the norm for quarterly macroeconomic instruments, where exogenous surprises explain only a small share of the variation in the endogenous variable, and the standard response applies here: I retain weak-instrument-robust Anderson–Rubin inference (Anderson and Rubin, 1949; Andrews et al., 2019) for the yield coefficient throughout. The direction of any remaining bias is also known. Weak instruments pull two-stage least squares toward ordinary least squares, and the simultaneity of demand pushes ordinary least squares downward, latent demand raises prices and lowers yields, so the estimated slopes are lower bounds and the estimated elasticities conservative. Because the first stage is common to every sector, the bias runs in the same direction everywhere and the cross-sector ranking is preserved. In practice the agreement across instrument families of different strengths, in Table A.11, indicates that the bias is small: severely attenuated estimates would differ across instruments of different strength, and they do not. Appendix D.7 reports the OLS benchmark directly: OLS understates the long-run slope of every sector type, most severely for the elastic sectors, mutual funds and the arbitrageurs, whose rebalancing is precisely the demand that moves yields, the direction and pattern simultaneity predicts.

Two independent instrument families corroborate the Treasury exclusion restriction. Table A.11 re-estimates the pooled Treasury slope using, first, only the foreign idiosyncratic demand shock, the most defensibly exogenous donor since foreign Treasury demand is reserve-driven (Warnock and Warnock, 2009), and second, the quarterly Jarociński and Karadi (2020) monetary-policy surprise, the instrument family of Eren et al. (2026), which is relevant on the post-2008 sample ($F = 15$). Each reproduces the granular-instrument baseline within one standard error, and the joint specifications do not reject the Sargan overidentification test ($p = 0.38$ with the foreign shock, $p = 0.06$ with the surprise). Three identification strategies, granular flows, reserve-driven foreign demand, and announcement-window surprises, converge on the same Treasury slope. The supply-shock instruments of the prior literature, the monetary-policy surprises of Swanson (2021), identify only the post-2010 sample and enter as a robustness check.

## 4.2 Rebalancing Speed

Figure 2 traces the horizon response $\beta_i^{(h)}$ of each sector type, and Table 4 reports the impact slope $\beta^{(0)}$ and the long-run slope $\beta^{(4)}$ for every sector type. I take the long-run slope as the cumulative response one year after the shock, $\beta^{(4)}$, a horizon fixed in advance by which the response has developed for every sector; the paths in Figure 2 are flat from $h \approx 4$ on. The long-term investors complete only a fraction of their rebalancing in the quarter of the shock. Insurers move about a third of the way on impact, $\beta^{(0)} = 0.04$, and reach $\beta^{(4)} = 0.09$, the most precisely estimated positive long-run slope in the system; banks show the same pattern at a similar scale, 0.03 to 0.07. Pension funds never arrive: their response is indistinguishable from zero at every horizon, with a point estimate that drifts slightly negative, consistent with allocations set by funding rules and mandates rather than by yields. The shapes of these

Table 3: **First-Stage $t$-Statistics**

| Asset | Instrument | Coefficient | First-stage $F$ | $N$ |
|---|---|---|---|---|
| Treasury | $\Gamma_{\text{Tre}}$ (granular) | $-2.397^{***}$ <br> $(-3.24)$ | 12.1 | 159 |
| Agency MBS | $\Delta\text{SOMA}_{\text{MBS}}$ | $-1.137^{***}$ <br> $(-2.90)$ | 11.0 | 138 |
| | $Z_{\text{MBS}}^{\text{Drukker}}$ | $-0.110$ <br> $(-1.08)$ | | |
| Corporate | $\text{Lewbel}_{\text{Corp}}$ (Maturity) | $-0.007$ <br> $(-0.16)$ | 7.3 | 159 |
| | $\text{Lewbel}_{\text{Corp}}$ (Issuance) | $0.095^{**}$ <br> $(2.19)$ | | |
| | $\text{Lewbel}_{\text{Corp}}$ (Coupon) | $-0.010$ <br> $(-0.26)$ | | |
| Municipal | $\text{Lewbel}_{\text{Muni}}$ (Maturity) | $-0.014$ <br> $(-0.28)$ | 7.6 | 159 |
| | $\text{Lewbel}_{\text{Muni}}$ (Issuance) | $0.193^{***}$ <br> $(2.94)$ | | |
| | $\text{Lewbel}_{\text{Muni}}$ (Coupon) | $-0.027$ <br> $(-0.54)$ | | |
| Joint $F$ (all instruments) | | | 8.5 | 615 |

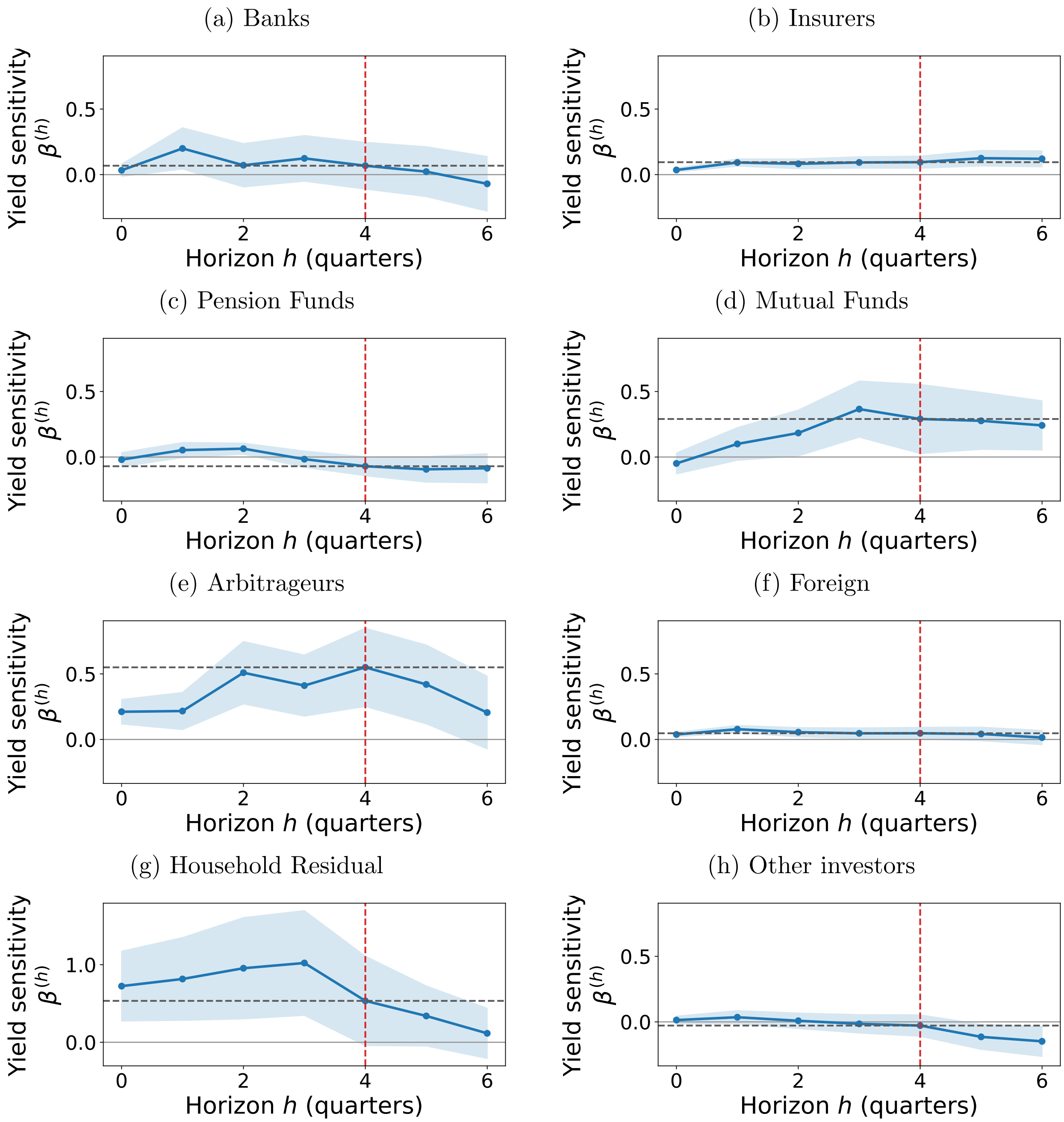
*Notes:* This table shows the first stage, estimated asset by asset on 1985Q1–2025Q1 quarterly data, with robust $t$-statistics in parentheses; the yield varies only at the asset–quarter level, so the first stage is common across sectors. Treasury yields are instrumented by the granular instrument $\Gamma$; agency MBS yields (option-adjusted, available from 1990Q3, hence $N = 138$) by the Fed's SOMA MBS purchases and the Drukker (2025) surprise; corporate and municipal yields by the Lewbel (2012) heteroskedasticity instruments, the demeaned quantity-side characteristic changes (maturity, log issuance, coupon) interacted with the first-stage residual, standardized to unit variance so the coefficients are comparable (the $F$ statistics are invariant to this scaling). All regressions control for the first-differenced asset characteristics (maturity, log issuance, coupon, and lagged duration). The first-stage $F$ tests each asset's own instruments jointly. The final row reports the joint $F$ of all nine instruments on the stacked asset $\times$ quarter panel with asset fixed effects; because the instrument matrix is block-diagonal, each instrument nonzero only on its own asset, this statistic averages strong and weak blocks, and the per-asset $F$ is the informative measure of each yield's identification. Anderson–Rubin inference is used for the yield coefficient throughout.

paths are estimated, not imposed. The insurer path is the geometric partial adjustment that van der Beck (2026) imposes for equity institutions, closing a roughly constant fraction of the remaining gap each quarter (Appendix D.6). The slow-moving capital of Duffie (2010) thus operates here at the scale of quarters, not years, and what the one-quarter response misses is less the timing than the level: it understates the long-run demand of insurers roughly threefold, which is why a static quarterly regression makes the market's largest long-term investors look inelastic.

The impact slopes reproduce the near-zero quarterly responses that a static differenced regression would find, and they explain why such a regression understates the market's absorption capacity. With the slow investors frozen at $\beta^{(0)} \approx 0$, the demand system understates how far investors ultimately rebalance, both within bonds and toward the outside asset; the long-run slopes restore it. The distinction matters for the counterfactual (Section 5): the impact response governs the initial dislocation, the long-run slope where yields settle.

The flow- and measurement-driven sectors, mutual funds, arbitrageurs, and households, are estimated on the 2010–2025 window, where their holdings are measurable: before 2010 the Financial Accounts residual that proxies for households is contaminated by hedge funds it cannot see, and broker-dealers operate under a pre-Dodd–Frank balance-sheet regime. The counterfactual uses the long-run slope $\beta^{(4)}$ for every sector, one uniform rule with no per-sector discretion. Two estimates deserve comment. Mutual funds reach $\beta^{(4)} = 0.29$, which partly reflects return-chasing investor flows arriving over the year rather than a pure demand slope; the uniform rule keeps it, and because an elastic sector absorbs supply in market clearing, doing so makes the QE and QT estimates conservative. Households are the Z.1 residual; their large impact slope $\beta^{(0)} = 0.72$ is a pooling artifact of the residual construction, not a behavioral elasticity: a per-asset decomposition places it entirely in the agency-MBS rows, where the household holding is itself the residual of the agency-pool accounts, while the Treasury rows, where the hedge funds reside, show no response at any horizon. Their long-run slope, 0.53 with a standard error of 0.58, is statistically indistinguishable from zero; the uniform rule keeps the point estimate, which, being elastic, again makes the counterfactual conservative. The arbitrageur sector reaches an elastic $\beta^{(4)} = 0.55$ that *damps* the impact as an elastic marginal absorber should (Vayanos and Vila, 2021). Foreign investors are inelastic at every horizon, consistent with a reserve-management mandate. Table 5 reports every slope with its estimation window.

Figure 2: **Sector Rebalancing Paths**




*Note:* Each panel plots the horizon response $\beta_i^{(h)}$ from the local projections of Equation (7),

$$\sum_{s=t}^{t+h} \Delta \ln\big[w_{is}(n)/w_{is}(0)\big] = \beta_i^{(h)}\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \tau_{in}\, t + u_{it}^{(h)}(n),$$

the cumulative rebalancing of a sector's duration-adjusted log bond share to a one-quarter yield change, with $\Delta y_t(n)$ instrumented by each asset's instrument set (Table 3). Bands are $\pm 1$ Newey–West standard error. The grey dashed line marks the long-run slope $\beta^{(4)}$ and the red dashed line the one-year horizon at which it is read. All other sectors are estimated on the full 1985–2025 sample; the flow- and measurement-driven sectors, mutual funds, arbitrageurs, and households, on 2010–2025. The counterfactual uses the long-run slope $\beta^{(4)}$ for every sector (Table 4).

Table 4: **Rebalancing Speed and Long-Run Slopes by Sector Type**

| Sector type | Impact $\beta^{(0)}$ | Long-run $\beta^{(4)}$ | Sample | $N$ |
|---|---|---|---|---|
| Banks | +0.03 | +0.07 | 1985–25 | 2,396 |
| | (0.05) | (0.18) | | |
| Insurers | +0.04 | +0.09 | 1985–25 | 1,198 |
| | (0.02) | (0.05) | | |
| Pension Funds | -0.02 | -0.07 | 1985–25 | 1,797 |
| | (0.05) | (0.07) | | |
| Mutual Funds | -0.05 | +0.29 | 2010–25 | 880 |
| | (0.08) | (0.26) | | |
| Arbitrageurs | +0.21 | +0.55 | 2010–25 | 440 |
| | (0.09) | (0.30) | | |
| Foreign | +0.04 | +0.05 | 1985–25 | 599 |
| | (0.02) | (0.05) | | |
| Household Residual | +0.72 | +0.53 | 2010–25 | 220 |
| | (0.45) | (0.58) | | |
| Other | +0.01 | -0.03 | 1985–25 | 5,391 |
| | (0.03) | (0.08) | | |

*Notes:* Sector-type local projections, Equation (7):

$$\sum_{s=t}^{t+h} \Delta \ln \frac{w_{is}(n)}{w_{is}(0)} = \beta_i^{(h)}\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \tau_{in}\, t + u_{it}^{(h)}(n),$$

with $\Delta y_t(n)$ instrumented by each asset's instrument set (Table 3). $\beta^{(0)}$ is the impact (quarterly) slope and $\beta^{(4)}$ the long-run slope, the cumulative response one year (four quarters) after the shock, with Newey–West (Bartlett kernel, bandwidth $h$) standard errors in parentheses. Driscoll–Kraay standard errors, which also allow cross-unit correlation within each quarter, leave every conclusion intact; the one estimate that widens, the insurer long-run slope, remains positive with $t = 1.6$ (Appendix D.5). All other sectors are estimated on the full 1985Q1–2025Q1 sample; the flow- and measurement-driven sectors (mutual funds, arbitrageurs, and households) on 2010Q1–2025Q1, the window given in the Sample column. $N$ is the number of institution–asset–quarter observations in the pooled horizon-4 projection. The counterfactual uses the long-run slope $\beta^{(4)}$ for every sector. Households are the Z.1 residual, whose large impact slope $\beta^{(0)} = 0.72$ is a pooling artifact of the residual construction rather than a demand elasticity (see the text).

Table 5 reports the impact and long-run slopes for all 26 individual sectors, each on its estimation window. The slopes are dispersed within each type, and foreign banking offices carry the steepest long-run slope among the banks. No individual long-run slope is estimated precisely, the weak-identification pattern that recurs in demand-system estimation and that motivates both the sector-type aggregation above and the robust inference I use throughout. I therefore read the demand system off the sector-type aggregates and treat the disaggregated slopes as descriptive.

### 4.3 Demand Elasticities

The long-run slopes imply the aggregate demand elasticity, which summarizes the demand inelasticity behind the counterfactual; the counterfactual itself clears the full asset-by-asset system. The post-2010 supply-shock specification of the prior demand-system literature, with its finer 28-sector split and its liquidity control, is reported as a specification check in Appendix D.4.

Figure 3 plots $\eta_{it}^{\text{bond}}$ from Equation (9) and the bond-AUM-weighted market elasticity, using the long-run slope $\beta^{(4)}$ for every sector, as in the counterfactual. The cross-sector ordering is stable throughout: the arbitrageurs, the market's fast marginal traders, and the household residual carry the highest measured elasticities, with mutual funds next, while banks, insurers, and foreign investors are inelastic at every date and pension funds essentially flat; the household series inherits the residual-construction caveat above. The pattern reflects each sector's slope and its bond share, since Equation (9) scales with $\beta_i\,(1-w_{it}^{\text{bond}})$: the elastic sectors combine sizable slopes with modest bond shares, so the outside-asset weight $(1-w)$ stays near one, while the inelastic sectors hold large bond positions at shallow slopes. The level moves with the yield environment. In the high-yield years of 1985–1990, when the market yield averaged 9%, the market elasticity was about 1.2; over the QE era of 2010–2021, with yields averaging 3%, it fell to about 0.3, and recovered to about 0.5 as yields rose in 2022–2025. Since 2010 aggregate bond demand has thus been well below one, consistent with the inelastic bond demand in the literature (e.g., Koijen and Yogo, 2019), and the more so because the largest holder, the foreign sector, is among the least elastic. Inelastic demand, a near-vertical demand curve in yield–quantity space, implies that the supply shifts induced

Table 5: **Estimated Demand Parameters by Sector**

| Sector | Demand coefficients | | | | | | |
|---|---|---|---|---|---|---|---|
| | Impact $\beta^{(0)}$ | Long-run $\beta^{(4)}$ | Maturity | Issuance | Duration | $N$ | Sample |
| (a) Banks | | | | | | | |
| Banks in affiliated areas | -0.07 | -0.20 | 0.03 | -0.38 | 0.03 | 615 | 1985–25 |
| | (-0.7) | (-1.0) | (0.03) | (0.80) | (0.04) | | |
| Credit unions | 0.02 | 0.41 | 0.01 | -0.35 | 0.16 | 615 | 1985–25 |
| | (0.3) | (1.5) | (0.02) | (0.47) | (0.15) | | |
| Foreign banking offices | 0.13 | 0.09 | -0.04 | -0.90 | -0.00 | 615 | 1985–25 |
| | (0.9) | (0.6) | (0.03) | (0.99) | (0.01) | | |
| U.S.-Chartered depositories | 0.04 | -0.02 | -0.01 | 0.45 | -0.01 | 615 | 1985–25 |
| | (1.2) | (-0.2) | (0.01) | (0.38) | (0.01) | | |
| (b) Insurers | | | | | | | |
| Life insurers | 0.02 | 0.10 | -0.02** | -0.03 | -0.01 | 615 | 1985–25 |
| | (1.0) | (1.3) | (0.01) | (0.14) | (0.01) | | |
| Property-Casualty insurers | 0.05*** | 0.09* | -0.02** | -0.01 | -0.01 | 615 | 1985–25 |
| | (2.6) | (1.7) | (0.01) | (0.14) | (0.01) | | |
| (c) Pension Funds | | | | | | | |
| Federal government funds | -0.18 | -0.13 | 0.04 | 0.31 | -0.09 | 615 | 1985–25 |
| | (-1.3) | (-0.7) | (0.06) | (0.70) | (0.06) | | |
| Private pension funds | 0.03 | 0.07 | -0.02* | -0.10 | -0.00 | 615 | 1985–25 |
| | (1.3) | (1.3) | (0.01) | (0.18) | (0.01) | | |
| State & local funds | 0.10** | -0.14 | -0.01 | 0.57* | -0.02 | 615 | 1985–25 |
| | (2.4) | (-1.2) | (0.01) | (0.33) | (0.02) | | |
| (d) Mutual Funds | | | | | | | |
| Closed-end funds | -0.03 | 0.07 | -0.01 | -0.74 | -0.00 | 236 | 2010–25 |
| | (-0.5) | (1.0) | (0.01) | (0.94) | (0.01) | | |
| Exchange-traded funds | -0.27** | 0.00 | 0.05*** | -4.91*** | -0.01 | 236 | 2010–25 |
| | (-2.3) | (0.0) | (0.02) | (1.60) | (0.01) | | |
| Money market funds | 0.12 | 0.68 | -0.01 | 1.21 | 0.02 | 236 | 2010–25 |
| | (0.6) | (0.9) | (0.04) | (3.51) | (0.02) | | |
| Mutual funds | -0.01 | 0.40 | -0.01 | -0.76 | -0.02 | 236 | 2010–25 |
| | (-0.1) | (1.4) | (0.02) | (1.37) | (0.02) | | |
| (e) Arbitrageurs | | | | | | | |
| mREITs | 0.21 | 0.95 | -0.01 | 1.91 | -0.02 | 236 | 2010–25 |
| | (1.5) | (1.5) | (0.04) | (2.18) | (0.02) | | |
| Brokers and dealers | 0.18 | 0.07 | -0.04* | 2.97 | 0.02 | 236 | 2010–25 |
| | (1.4) | (0.3) | (0.02) | (2.29) | (0.01) | | |

Table 5: **Estimated Demand Parameters by Sector** (continued)

| Sector | Demand coefficients: Impact $\beta^{(0)}$ | Long-run $\beta^{(4)}$ | Maturity | Issuance | Duration | $N$ | Sample |
|---|---|---|---|---|---|---|---|
| (f) Foreign | | | | | | | |
| Rest of the world | 0.04** | 0.05 | -0.01** | 0.19* | -0.01 | 615 | 1985–25 |
| | (2.2) | (1.0) | (0.01) | (0.10) | (0.01) | | |
| (g) Households | | | | | | | |
| Households | 0.72 | 0.53 | -0.18** | 7.77 | 0.11 | 236 | 2010–25 |
| | (1.6) | (0.9) | (0.08) | (6.31) | (0.07) | | |
| (h) Other | | | | | | | |
| Agency/GSE pools | 0.06*** | 0.03*** | -0.01*** | -0.01 | 0.00 | 615 | 1985–25 |
| | (35.4) | (2.8) | (0.00) | (0.02) | (0.00) | | |
| Federal government | 0.04 | -0.57 | -0.06 | 0.32 | 0.10 | 615 | 1985–25 |
| | (0.9) | (-1.6) | (0.05) | (0.44) | (0.06) | | |
| Finance companies | 0.07*** | 0.06* | -0.01* | 0.09 | 0.00 | 615 | 1985–25 |
| | (5.9) | (1.9) | (0.00) | (0.11) | (0.00) | | |
| GSEs | -0.07 | -0.21 | 0.01 | 0.66 | -0.01 | 615 | 1985–25 |
| | (-0.5) | (-0.7) | (0.02) | (0.67) | (0.02) | | |
| Holding companies | 0.04 | 0.17 | -0.03 | -0.16 | 0.02 | 615 | 1985–25 |
| | (0.6) | (1.1) | (0.03) | (0.36) | (0.02) | | |
| ABS issuers | 0.13 | 0.22 | -0.01 | 0.76 | 0.03 | 615 | 1985–25 |
| | (1.4) | (0.7) | (0.05) | (0.80) | (0.03) | | |
| Nonfinancial business | 0.03 | -0.09 | 0.00 | -0.05 | 0.02 | 615 | 1985–25 |
| | (1.0) | (-0.9) | (0.02) | (0.17) | (0.01) | | |
| Other financial business | -0.22 | 0.11 | 0.08 | -1.55 | -0.02 | 615 | 1985–25 |
| | (-0.9) | (0.2) | (0.07) | (2.07) | (0.02) | | |
| State & local governments | 0.07*** | 0.04 | -0.01* | 0.34** | 0.01** | 615 | 1985–25 |
| | (4.2) | (0.9) | (0.01) | (0.15) | (0.01) | | |

*Notes:* Demand coefficients for each of the 26 sectors, grouped by sector type, each estimated on the window given in the final column: the impact response $\beta^{(0)}$ and the one-year response $\beta^{(4)}$ from the sector-level local projections, Equation (7) estimated sector by sector,

$$\sum_{s=t}^{t+h} \Delta \ln \frac{w_{is}(n)}{w_{is}(0)} = \beta_i^{(h)}\, \Delta y_t(n) + \gamma_i' \Delta x_t(n) + \nu_{in}' \Delta W_t + \tau_{in}\, t + u_{it}^{(h)}(n),$$

with $\Delta y_t(n)$ instrumented by each asset's instrument set (Newey–West $t$-statistics in parentheses). All other sectors use the full 1985Q1–2025Q1 sample; the flow- and measurement-driven sectors, mutual funds (return-chasing flows), the arbitrageurs (broker-dealers and mREITs, whose balance sheets operate under the post-Dodd–Frank regime), and households (the Z.1 residual, unreliable before 2010), use 2010Q1–2025Q1. The counterfactual reads demand off the sector-type aggregates of Table 4, using $\beta^{(4)}$ for every sector; the unit-level slopes here are the disaggregated detail behind those aggregates. The maturity, issuance, and duration coefficients are from the first-difference IV regression on the same window, with robust standard errors in parentheses. The slopes are imprecise at this disaggregation, the reason I read the demand system off the sector-type aggregates and use Anderson–Rubin inference. *, **, and *** denote $|t|$ above 1.64, 1.96, and 2.58.

by QE move yields by large amounts. The horizon matters as well. Insurers and banks are roughly two- to threefold more elastic in the long run than on impact, the horizon gap that van der Beck (2026) documents for equity institutions, though banks and insurers close the gap within a quarter or two; mutual-fund flows and the arbitrageurs' full response arrive over the year. Because these sectors together hold most of the market, the aggregate demand curve is markedly more inelastic on impact than in the long run, so a one-off balance-sheet shock moves yields on impact roughly three times as much as once slow capital has arrived, the impact overshoot traced in Section 5. This is the horizon duality of elasticities and price impacts that van der Beck (2026) formalizes; here the counterfactual clears the full nonlinear system with the supply response of Section 5.2, which absorbs part of the wedge.

This yield dependence is structural, $\eta_{it}^{\text{bond}} \propto \bar{y}_t$, and reading it correctly matters for the QE–QT comparison. The elasticity compares proportional changes, while investors respond to basis points: a one-percent move in yields carries a fifth of the basis points at the 2 percent yields of the QE era that it carries at 10 percent, so demand measured per percent responds less. In market clearing the level cancels. The semi-elasticities $\beta_i$ price the absorption of a supply shift in basis points, so the impact of removing a given fraction of the float is pinned down by the yield-free slope $\sum_i s_{it}\, \beta_i\, (1 - w_{it})$, whatever the level of yields; low yields do not, by themselves, make a given balance-sheet change move yields further. What the level governs is the proportional repricing those basis points represent, which is what the falling elasticity records. Appendix B.1 makes the attribution exact: the yield level accounts for 93.8 percent of the quarterly variation in the market elasticity, while the yield-free slope component is essentially flat over the sample, so the QE-era decline in the elasticity is the yield level, not a change in the composition of holdings. That aggregate slope is nearly flat, but who holds a given asset's float still varies: the Treasury float has shifted toward the elastic sectors since the QE era, which is why the QT episodes move yields somewhat less per unit of float (Section 5). Figure 3, Panel (b) plots the resulting price impact per 1 percent of float, asset by asset. The Treasury impact peaks near 23 basis points in the mid-

2000s, when the foreign official sector, the largest and least elastic holder, was accumulating Treasuries, and falls to about 7 basis points by 2025 as the float shifts toward the elastic sectors; the municipal impact sits near 3 basis points throughout, because households, an elastic holder, own most of the float. This series, not the elasticity, is the demand-side capacity that balance-sheet policy runs against, and its movements are composition alone.

The demand system also implies a full set of own- and cross-yield elasticities across the four bond markets. Appendix Figure A.7 plots their market-level time series: each panel fixes a one-percent rise in one asset's yield and traces the response of demand for all four assets. The own-yield elasticities (solid) are sizable and move with the yield environment, falling by more than half from the high-yield 1980s to the QE era and recovering as yields rose after 2022; the cross-yield elasticities (dashed) sit near zero throughout, an order of magnitude smaller, because each bond is a small share of investors' portfolios. Demand-side substitution thus runs mainly between bonds and the non-bond outside asset, not across bonds; the counterfactual spillovers of Section 5 are the general-equilibrium counterpart, in which each market re-clears.

# 5 Counterfactual Simulation

In this section, I simulate the U.S. bond market equilibrium under counterfactual scenarios, focusing on the counterfactual QE and QT.

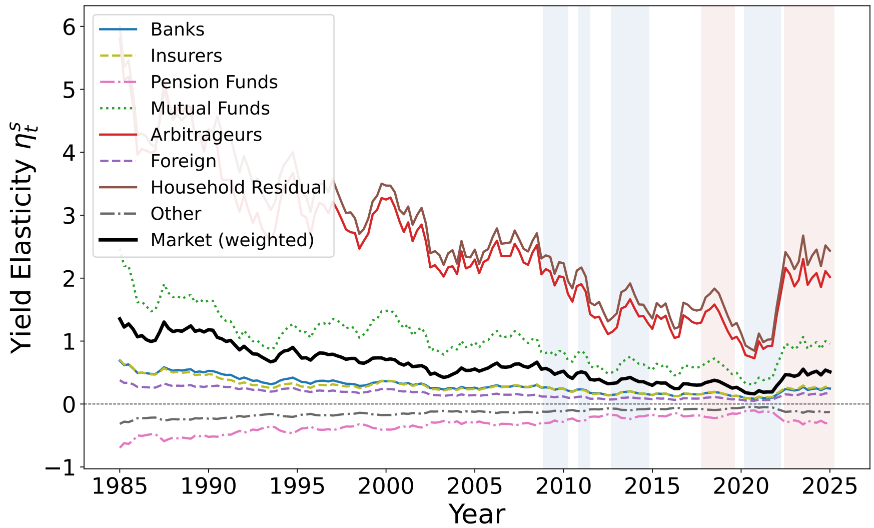
## 5.1 Setup

The equilibrium asset yields are jointly determined by the estimated demand system in (4) and the market clearing condition in (11). To close the asset pricing model, the market for each asset $n$ clears in each period $t$:

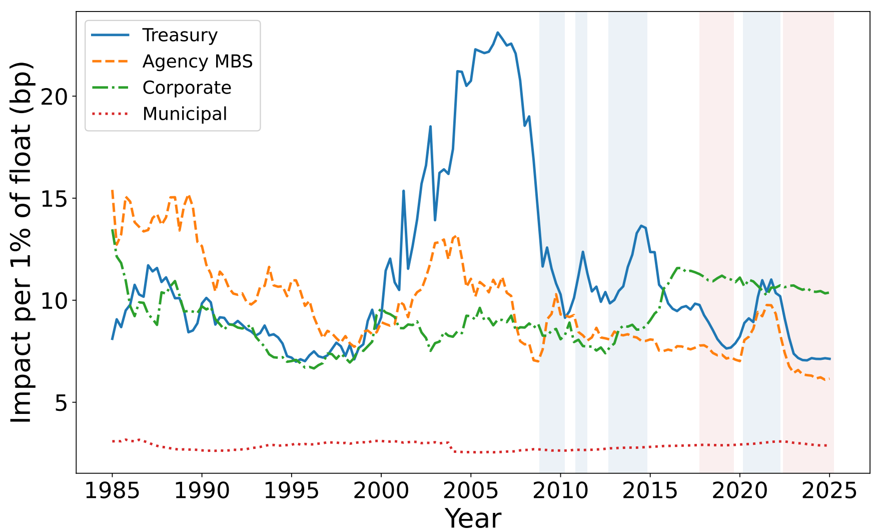
$$\sum_{i=1}^{I} w_{it}^{*}(\boldsymbol{y}_t^{*},\, \boldsymbol{x}_t,\, \boldsymbol{\beta}_i,\, \boldsymbol{\varepsilon}_{it}) \cdot A_{it} = S_t(n;\, \boldsymbol{y}_t^{*}), \tag{17}$$

Figure 3: **Bond Demand Elasticity and Absorption Capacity**

(a) Sector-level and aggregate bond demand elasticity




(b) Price impact per 1 percent of float




*Note:* Panel (a) plots the bond demand elasticity $\eta_{it}^{\text{bond}} = \beta_i\,(1 - w_{it}^{\text{bond}})\,\bar{y}_t$, Equation (9), for each sector type; the aggregate market elasticity is the bond-AUM-weighted average. The arbitrageur aggregate comprises broker-dealers and mREITs throughout; the Form-PF hedge-fund unit, available only from 2012 and without a reported demand coefficient, is excluded so the sector's composition is constant across the sample. Panel (b) plots the price impact of Equation (12), $1/\sum_i s_{it}(n)\,\beta_i\,(1-w_{it}(n))$, the basis-point yield impact of removing 1 percent of asset $n$'s private float, where $s_{it}(n)$ is institution $i$'s share of the float (which excludes the Federal Reserve) and $w_{it}(n)$ is its own portfolio share; the series is the first-order linearization of the market-clearing counterfactual of Section 5 and carries no yield term, so unlike Panel (a) its movements reflect the composition of holdings alone. Both panels use the counterfactual demand slopes ($\beta^{(4)}$ for every sector, with the 2010-window slopes for mutual funds, the arbitrageurs, and the household residual). Blue bands mark the four QE episodes, red bands the two QT episodes.

where $\boldsymbol{y}_t^*$ is the equilibrium yield vector, $\boldsymbol{x}_t$ is the asset-characteristics vector, $\boldsymbol{\beta}_i$ are the estimated demand parameters, $\boldsymbol{\varepsilon}_{it}$ are the latent demand shocks, and $S_t(n;\, \boldsymbol{y}_t^*)$ is the supply schedule of asset $n$, $S_t(n;\, \boldsymbol{y}) = f_n\big(\boldsymbol{y};\, \bar{S}_t(n)\big)$, which lets net issuance respond to the equilibrium yield; $\bar{S}_t(n)$, the free float at baseline yields, is the schedule's shifter, and at baseline yields supply equals the observed float. Section 5.2 estimates the schedule's reduced form. This system comprises $N \times T$ equations, and the equilibrium yield vector $\boldsymbol{y}_t^*$ is the implicit function:

$$\boldsymbol{y}_t^* = \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t\big), \tag{18}$$

where $\boldsymbol{g}(\cdot)$ is the implicitly defined equilibrium mapping, which clears every market against the supply schedule, $\bar{\boldsymbol{S}}_t$ is the vector of baseline floats, the schedule's shifter, and $\boldsymbol{A}_t$ is the vector of investor assets under management.

I evaluate the impact of counterfactual monetary policy by comparing baseline and counterfactual equilibrium yields. Changes in the Fed's asset holdings translate into changes in the baseline private free float $\bar{\boldsymbol{S}}_t$. This one-for-one translation holds the fiscal path fixed: debt the Fed does not roll over is refinanced to the public at auction, and agency MBS paydowns are re-securitized, so runoff adds to the float exactly as outright sales would; any yield-induced change in gross issuance is what the supply schedule captures. When the Fed purchases agency MBS, that quantity is withdrawn from the public market, reducing the supply available to private investors; the issuance response to the counterfactual yield then operates through the supply schedule inside $\boldsymbol{g}(\cdot)$. Denoting the post-purchase baseline float by $\bar{\boldsymbol{S}}_t^{\mathrm{CF}}$, the yield effect is

$$\Delta \boldsymbol{y}_t = \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t^{\mathrm{CF}}\big) - \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t\big). \tag{19}$$

## 5.2 Supply Response

The counterfactual lets bond supply respond to yields: when the counterfactual yield rises, issuers issue less (Fang and Xiao, 2025), attenuating the yield effect. This supply side is deliberately reduced-form, a first-order net-issuance elasticity rather than a structural model of issuer behavior, since the demand system is the paper's structural focus. I estimate these elasticities from the same Flow-of-Funds data as the demand system. The identification mirrors the demand side. Estimating a demand slope requires supply-side variation in yields; estimating the supply (issuance) response requires *demand*-side variation, and the paper's instruments, the granular instrument, the Drukker (2025) surprise, and the Lewbel instruments, are exactly that: none is driven by issuers. The specification parallels the local projections of the demand system, with the response read at the same one-year horizon $h = 4$ as the demand slope $\beta^{(4)}$: the cumulative growth of outstanding through one year after a one-quarter yield change,

$$\sum_{s=t}^{t+4} \Delta \ln S_s(n) \times 100 = \kappa_i^{(4)} \, \Delta y_t(n) + \text{quarter fixed effects} + \tau \, t + u_t(n), \tag{20}$$

with $\Delta y_t(n)$ instrumented, quarter fixed effects absorbing the Treasury refunding cycle, and the 2003Q4–2004Q1 statistical revision of municipal outstanding removed. The implied supply elasticity is $\psi_n = -\kappa_n^{(4)}$, the percent of outstanding withdrawn per percentage-point yield rise per year. The counterfactual's supply schedule is its first-order form,

$$S_t(n; \boldsymbol{y}) = \bar{S}_t(n) \left[1 - \psi_n \, \Delta y_t(n)\right], \tag{21}$$

so relative to the baseline float $\bar{S}_t(n)$, issuers withdraw $\psi_n$ percent of outstanding per percentage point that the counterfactual yield rises above its baseline value.

Table 6 reports the estimates next to the Fang and Xiao (2025) values. Every asset carries the expected negative sign. Corporate issuance is the most precisely estimated and the most responsive ($-5.5$, $t = -2.5$), the cross-market issuance timing of Ma (2019): corporations accelerate issuance when yields fall. Treasury issuance is essentially unresponsive ($-0.2$, standard error 5.9), consistent with issuance driven by fiscal needs rather than rates. Agency MBS ($-1.8$) sits within one standard error of the Fang and Xiao (2025) value (2.2), and the municipal elasticity, which they do not cover, is right-signed but imprecise ($-1.6$, standard error 3.9). The counterfactual uses the own estimates; results with the Fang and Xiao (2025) values are similar in pattern.

Table 6: **Net-Issuance Elasticities Estimated from the Flow of Funds**

| | One-year issuance response $\kappa^{(4)}$ | | Implied $\psi$ | Fang–Xiao |
|---|---|---|---|---|
| Asset | Coefficient | (s.e.) | (%/pp/yr) | (%/pp/yr) |
| Treasury | $-0.20$ | (5.87) | 0.20 | 0.96 |
| Agency MBS | $-1.84$ | (2.88) | 1.84 | 2.24 |
| Corporate | $-5.45^{**}$ | (2.16) | 5.45 | 1.67 |
| Municipal | $-1.60$ | (3.86) | 1.60 | — |

*Notes:* Instrumental-variables local projections of cumulative outstanding growth (percent) over the year following a one-quarter yield change, per asset, on 1985Q1–2025Q1; Newey–West standard errors in parentheses. Instruments are demand-side: the granular instrument for Treasuries, the Drukker (2025) surprise for agency MBS (SOMA purchases are excluded because they shift the private float, not gross issuance), and the Lewbel instruments for corporate and municipal yields. The implied $\psi$ is the elasticity used in the counterfactual's supply response. The final column reports the values of Fang and Xiao (2025) (Table A3), who do not cover municipal bonds.

## 5.3 Assessment of Historical QE/QT Episodes

I apply the structural counterfactual to the four QE programs (QE1–QE4) and the two QT episodes (QT1, 2017–2019; QT2, 2022–2025). The extended sample brings QE1 in-sample, the episode prior demand-system studies omit for lack of pre-2010 data. For each episode I hold investor characteristics, long-run demand slopes, and AUM at their observed values, let bond supply respond to yields through net-issuance elasticities estimated from the same Flow-of-Funds data (Section 5.2), and ask how yields would differ had the Fed held its portfolio at the pre-episode level. Table 7 reports the long-run effects, the response once every sector has completed rebalancing; absent the supply response they are materially larger.[7]

Five findings stand out. First, QE1 worked mainly through mortgage markets. It was the Fed's only MBS-heavy program, \$1.2 trillion of agency MBS against \$300 billion of Treasuries, and the counterfactual compresses MBS yields by 119 basis points against 50 for Treasuries. The MBS figure is of the same order as the event-study estimates of Hancock and Passmore (2011) (−85) and Krishnamurthy and Vissing-Jorgensen (2011) (−107): for a largely unanticipated program whose flow purchases nonetheless ran for over a year, the announcement window and the full-stock counterfactual measure similar objects, the window adding the signaling channel and the counterfactual adding the unanticipated tail of the purchases.

Second, the composition of purchases matters as much as their scale. QE2 compressed Treasury yields by 108 basis points but pushed Agency MBS yields *up* by 17, because the Fed ran off \$207 billion of MBS even as it bought Treasuries, adding to the MBS supply the private sector had to hold. Its narrow asset mix also muted spillovers, with Corporate and Municipal yields down 2 to 4 basis points.

Third, QE3 and QE4 combined Treasury and MBS purchases and compressed both markets, by 94 and 74 basis points under QE3 and by 183 and 107 under QE4. The broader mix widened the spillovers: Corporate and Municipal yields fell by up to 11 basis points, about three times the QE2 figures. QE4 is the largest program because its shock was largest, not because demand was steeper. It removed roughly a fifth of the Treasury free float, and scaled by that shock it moved yields *least* of the QE episodes, 8.8 basis points per 1 percent

[7] The arbitrageur sector, broker-dealers and mortgage REITs, is the bond market's elastic, fast-moving capital, the marginal absorbers in the spirit of Vayanos and Vila (2021); I assign it the elastic long-run slope estimated on the post-2010 window, on which the dealer balance sheet operates under the current regulatory regime. Because mortgage REITs hold agency MBS, this elasticity matters most there, consistent with the micro evidence of Frame and Steiner (2022), who show that agency-mREIT asset growth moves inversely with the Fed's MBS purchases: setting the arbitrageur slopes to zero, as Eren et al. (2026) do for their arbitrageur sector, raises the agency-MBS estimates and leaves the Treasury estimates little changed. Hedge funds, whose Treasury positions reflect the cash–futures basis trade rather than yield demand, are carved out for measurement and carry no demand parameter (Appendix D.1).

of float (final column), against 10 to 12 for QE1 to QE3.

Fourth, the spillover reaches Corporate and Municipal markets in similar measure to each other, a shift from settings that measure demand within the bond universe alone. The cross-asset effect scales with the product of portfolio weights, and the sectors that hold large Treasury positions also hold sizable Corporate and Municipal allocations, so rebalancing pressure reaches both markets. The underlying demand-side cross-elasticities are small, an order of magnitude below the own-yield elasticities (Appendix Figure A.7), so the spillovers reflect the size of the supply shocks rather than strong bond-versus-bond substitution, and the spillovers themselves are an order of magnitude below the own-market effects. In this sense the transmission is narrow, in the spirit of Krishnamurthy et al. (2013) and of the partially segmented markets of Greenwood et al. (2018), where specialists absorb local supply and cross-market capital arrives only gradually: balance-sheet policy concentrates its impact in the markets the Fed buys, and a broad duration channel, under which purchases in one market would move all long-term yields alike, finds little support. The rebalancing channel itself is real, and the micro evidence that QE-exposed mutual funds shift into corporate bonds (Selgrad, 2023) identifies exactly the elastic rebalancers of this system; the demand system adds that their aggregate footprint on non-purchased yields, while measurable, is an order of magnitude below the own-market effect. The narrowness is a statement about class-level totals, not about every pair of securities: with security-level data, Fang and Xiao (2025) recover substitution the four-class system averages over, such as between long-maturity Treasuries and agency MBS, close substitutes at matched duration, so purchases concentrated in one duration segment can reach the matched segment of a neighboring class more strongly than the class-level spillover suggests. The spillover is largest under the broad-mix programs QE3 and QE4, where two large markets are shocked at once.

Fifth, the QT episodes reverse the sign. QT1 unwound \$285 billion of Treasuries and \$300 billion of Agency MBS, raising the two yields by 16 and 23 basis points; QT2 unwound \$2.0 trillion and \$0.75 trillion, raising them by 69 and 44, with Corporate and Municipal yields up about 3 each. Per unit of float, the QT episodes moved yields less than any QE program (8.3 and 7.9 basis points per 1 percent of float against 8.8 to 12.0), reflecting the composition of the float at the time: by 2017–2019 and 2022–2025, holdings had shifted toward the elastic sectors, so each unit of supply met a flatter demand curve (Figure 3, Panel (b)). The yield level plays no role in these basis-point impacts; it cancels in market clearing (Section 4).

For the Treasury-heavy programs these impacts exceed announcement-window event studies by a factor of two to three, while for QE1 they sit below an event-study record that mixes in the signaling channel; the differences reflect the size and anticipation of the shocks, not an outsized demand slope. Scaled by the shock, the Treasury estimates run 8 to 12 basis points per 1 percent of float (final column of Table 7), squarely in the range of other demand-system estimates: Eren et al. (2026) report 8 to 13 basis points per percentage point of the central bank's share, and Chaudhary et al. (2026) 13 basis points per 1 percent shift in demand. The per-unit impact is also remarkably uniform across episodes, 8 to 12 for programs whose raw effects span 16 to 183 basis points: the headline numbers track the size of the shocks, not movements in the demand slope. Announcement windows, by contrast, price only the surprise component of purchases, and markets anticipated most QE2 and QE3 purchases well before the FOMC confirmed them. Event-study responses also include the signaling channel, which my supply-only counterfactual excludes: Krishnamurthy and Vissing-Jorgensen (2011) attribute 20 to 40 of QE1's 107 basis points to signaling. The full-stock estimates therefore exceed announcement effects even though they capture a narrower set of channels.

The QE4 figure illustrates that the headline number tracks the shock, not the slope. Two-thirds of QE4's $3.5 trillion of Treasury purchases, $2.3 trillion, came in the first half of 2020, when the Fed bought in weeks to restore market functioning rather than to compress term premia (Vissing-Jorgensen, 2021). Re-basing the counterfactual to the stock-QE phase that followed isolates a $1.0 to $1.2 trillion program that moved Treasury yields by 62 to

Table 7: **Yield Effects of Historical QE/QT Episodes**

(a) Quantitative easing

| Episode | Asset | ΔSOMA ($B) | $\Delta Y$ (bp) | $\lvert\Delta Y\rvert$/1% Float |
|---|---|---|---|---|
| QE1 (2008Q4–2010Q1) | Treasury | 300 | -49.9 | 11.3 |
| | Agency MBS | 1,223 | -119.0 | 6.5 |
| | Corporate | — | -7.1 | — |
| | Municipal | — | -5.1 | — |
| QE2 (2010Q4–2011Q2) | Treasury | 808 | -108.0 | 10.2 |
| | Agency MBS | -207 | 17.5 | 5.8 |
| | Corporate | — | -3.7 | — |
| | Municipal | — | -2.4 | — |
| QE3 (2012Q3–2014Q4) | Treasury | 772 | -93.8 | 12.0 |
| | Agency MBS | 845 | -73.6 | 5.7 |
| | Corporate | — | -5.0 | — |
| | Municipal | — | -3.7 | — |
| QE4 (2020Q1–2021Q4) | Treasury | 3,511 | -182.7 | 8.8 |
| | Agency MBS | 1,210 | -107.0 | 7.4 |
| | Corporate | — | -10.9 | — |
| | Municipal | — | -8.0 | — |

(b) Quantitative tightening

| Episode | Asset | ΔSOMA ($B) | $\Delta Y$ (bp) | $\lvert\Delta Y\rvert$/1% Float |
|---|---|---|---|---|
| QT1 (2017Q4–2019Q3) | Treasury | -285 | 16.5 | 8.3 |
| | Agency MBS | -300 | 22.6 | 6.2 |
| | Corporate | — | 1.4 | — |
| | Municipal | — | 1.1 | — |
| QT2 (2022Q2–2025Q1) | Treasury | -2,019 | 69.3 | 7.9 |
| | Agency MBS | -749 | 43.8 | 5.7 |
| | Corporate | — | 3.0 | — |
| | Municipal | — | 3.3 | — |

*Notes:* Yield effects (basis points) of holding the Fed's portfolio at its pre-episode level, from the market-clearing counterfactual of Section 5. The counterfactual shifts the baseline float by the Fed's balance-sheet change, $\bar{\boldsymbol{S}}_t^{\mathrm{CF}} = \bar{\boldsymbol{S}}_t + \boldsymbol{\Delta}\mathrm{SOMA}_t$, where ΔSOMA is the change in Fed holdings over the episode, and re-solves the equilibrium against the supply schedule of Equation (21), with the net-issuance elasticities $\boldsymbol{\psi}$ of Table 6 accumulated over the episode horizon. All columns are measured at the episode-end quarter; dashes mark assets the Fed did not purchase. The positive Agency MBS entry in QE2 reflects the Fed's concurrent MBS runoff. Corporate and Municipal entries are cross-asset rebalancing spillovers only. The final column scales the impact by the size of the supply shock: $\lvert\Delta Y\rvert$ per 1 percent of the private free float removed from (QE) or returned to (QT) the market.

76 basis points (Table A.12), comparable to QE3, while the per-unit impact holds at 9 to 11 basis points throughout. Attributing the whole \$3.5 trillion to a stock effect overstates the program's yield effect; the demand slope behind it is stable. Three caveats bound the interpretation, all in the same direction. The counterfactual holds latent demand and AUM fixed and treats the supply change as permanent, so the long-run estimates are an upper bound if investors expected the Fed to unwind (Jansen et al., 2024); any weak-instrument bias works the same way, since it pulls the demand slopes downward and a steeper demand curve raises the implied yield effect; and the demand slopes are identified from local variation, while the episodes shift supply well outside that range. The estimates are best read as upper bounds on the stock effects.

Figure 4 places the estimates in the literature, episode by episode. In the Treasury panel, my estimates sit closest to the other demand-system estimates. The model-implied per-\$100 billion impacts of Jansen et al. (2024), 3 to 14 basis points on the 10-year yield depending on the perceived persistence of the program, overlap the per-unit effects estimated here, 6 to 12 basis points per \$100 billion for the large programs and higher for the small early ones; they evaluate generic purchase impulses rather than the historical programs, so they enter this comparison per unit rather than per episode. Eren et al. (2026), who identify demand from monetary-policy surprises rather than supply shocks, report 113 basis points for QE2 against my 108, 57 for QE3 against my 94, 43 for QT1 against my 16, and 80 for QT2 against my 69: the two systems agree closely on QE2 and QT2, while mine reads QE3 larger and QT1 smaller. QE1, which the post-2010 demand systems cannot reach, enters here at 50 basis points, between the calibrated preferred-habitat estimate of 25 to 30 and the top of the term-structure range of 29 to 52, and below the event-study estimates of 91 to 107 that mix in the signaling channel. The horizon duality of Section 4 adds a second wedge: an announcement window is priced by the fast movers against the impact demand curve, roughly three times steeper than the long-run curve, so even a clean surprise can overstate the permanent effect. The post-announcement decay of QE effects that Wright

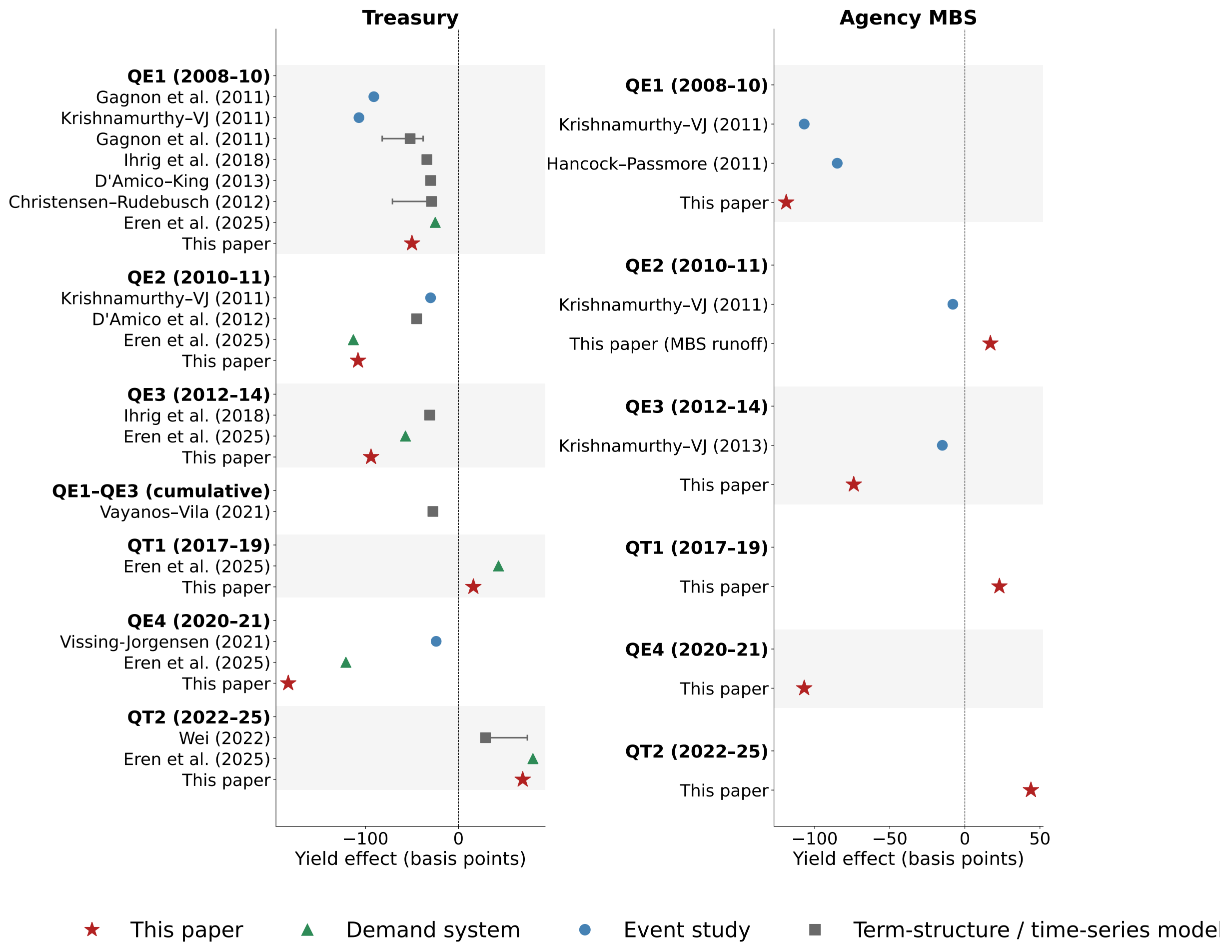
(2012) documents is, in this reading, the arrival of slow capital closing the overshoot, not evidence against the stock effect. The calibrated preferred-habitat benchmark of Vayanos and Vila (2021) anchors the bottom of the structural range: their 25 to 30 basis points for the cumulative QE1 to QE3 purchases assumes the demand shift mean-reverts, whereas my counterfactual treats the stock removal as permanent. The distance between these two structural estimates thus measures the role of perceived persistence, the margin Jansen et al. (2024) emphasize.

The Agency MBS panel is sparser because prior estimates are. Event studies cover QE1, where announcements lowered current-coupon MBS yields by 107 basis points and mortgage rates by 85 (Krishnamurthy and Vissing-Jorgensen, 2011; Hancock and Passmore, 2011), and QE3, where the announcement moved MBS yields by only 15 basis points (Krishnamurthy et al., 2013). For QE1 my estimate of 119 basis points is of the same order as the 85 mortgage-rate response of Hancock and Passmore (2011) and the 107 of Krishnamurthy and Vissing-Jorgensen (2011), measured on the option-adjusted MBS yield the demand system prices. My QE3 estimate of 74 basis points is several times the announcement response, because the open-ended flow purchases were largely anticipated and little of their stock effect appears in the announcement window. The QE2 entries differ even in sign. The two-day window shows MBS yields falling by 8 basis points, which Krishnamurthy and Vissing-Jorgensen (2011) attribute to signaling, while my supply-only counterfactual attributes a 17 basis point increase to the Fed's concurrent MBS runoff; the two numbers measure different channels rather than contradicting each other. For QE4 and the two QT episodes, prior MBS estimates are scarce, and the demand system provides structural numbers for episodes the event-study literature has not covered.

## 5.4 Prospective QT Scenario Analysis

I next project the yield effects of further quantitative tightening. Fixing the 2025Q1 portfolio and running additional QT at the QT2 Treasury:MBS pace, I consider three scenarios: a mild

Figure 4: **Historical QE/QT Yield Effects: This Paper versus Prior Estimates**




*Note:* This figure compares the structural full-stock estimates of Table 7 (stars) with prior estimates of the yield effect of each QE/QT episode, for Treasuries (left) and agency MBS (right). Circles are announcement-window event studies (Gagnon et al., 2011; Krishnamurthy and Vissing-Jorgensen, 2011; Krishnamurthy et al., 2013; Hancock and Passmore, 2011; Vissing-Jorgensen, 2021); squares are term-structure and time-series estimates (Gagnon et al., 2011; D'Amico et al., 2012; D'Amico and King, 2013; Christensen and Rudebusch, 2012; Ihrig et al., 2018; Wei, 2022); triangles are demand-system estimates (Eren et al., 2026). Horizontal bars show reported ranges: for Christensen and Rudebusch (2012), the term-premium component of the QE1 response across their models; for Wei (2022), normal versus turbulent conditions. The Vayanos and Vila (2021) entry is a preferred-habitat calibration of the cumulative QE1–QE3 purchases (12 percent of GDP) and appears as its own group. Episode definitions differ slightly across studies; Eren et al. (2026) date episodes by changes in the central bank's share of the market.

$400 billion, a baseline $1 trillion, and an aggressive $1.6 trillion of further runoff. Table 8 reports the results. A further $1 trillion of QT raises Treasury yields by 24 basis points and Agency MBS yields by 18 basis points, with Corporate and Municipal yields up about 1 to 2 each; the aggressive $1.6 trillion scenario raises Treasury and MBS yields by 39 and 26 basis points. A back-of-the-envelope calculation conveys the real stakes. Applied to the roughly $27 trillion of marketable federal debt as it reprices, the baseline Treasury rise adds about $65 billion a year to federal interest costs, and $104 billion under the aggressive scenario, a recurring fiscal cost to weigh against the balance-sheet reduction. On the household side, agency MBS yields pass through nearly one for one to primary mortgage rates, so the baseline adds about $47 a month to the payment on a newly originated $400,000 thirty-year fixed-rate mortgage, and the aggressive scenario about $70.[8] The marginal Treasury impact is nearly constant, about 2.4 basis points per $100 billion of additional runoff, because Treasury issuance barely responds to yields; the supply offset that would flatten it operates mainly in the corporate and MBS markets. The household residual is the projection's most consequential input: its long-run slope, 0.53 with a standard error of 0.58, is statistically indistinguishable from zero, and setting it to zero raises the baseline $1 trillion Treasury estimate from 24 to 45 basis points, and more than triples the municipal spillover, whose float households dominate. The baseline keeps the elastic point estimate under the uniform rule, the conservative choice, so the projections are best read as a lower bound with respect to household absorption. These are long-run effects; Figure 5 traces how they arrive.

Figure 5 projects the resulting yield path, combining the runoff pace with the estimated rebalancing dynamics. Each quarter's incremental runoff first lands on the sectors that respond at once: on impact it moves Treasury yields roughly three times their long-run effect, the overshoot obtained by re-solving the QT2 equilibrium horizon by horizon as the slow sectors travel from $\beta^{(0)}$ to $\beta^{(4)}$, and the excess dissipates within a year as mutual-fund

[8] Fiscal: $27 trillion × 24.0 (38.7) basis points ≈ $65 ($104) billion per year once the stock has repriced. Mortgage: the monthly payment on a loan of size $P$ at annual rate $r$ is the standard annuity $M(r) = P\,(r/12)\,/\,[1-(1+r/12)^{-360}]$; with $P$ = $400,000 and $r = 6.5\%$, $M(r+17.5\text{bp}) - M(r) \approx \$47$ and $M(r+26.1\text{bp}) - M(r) \approx \$70$ per month.

Table 8: **Yield Effects of Prospective QT Scenarios**

| Scenario | ΔSOMA Tsy/MBS ($B) | Tsy (bp) | MBS (bp) | Corp (bp) | Muni (bp) |
|---|---|---|---|---|---|
| Mild ($400B) | −292/−108 | 9.5 | 7.6 | 0.8 | 0.6 |
| Baseline ($1T) | −729/−271 | 24.0 | 17.5 | 1.4 | 1.3 |
| Aggressive ($1.6T) | −1,167/−433 | 38.7 | 26.1 | 1.7 | 2.0 |

*Notes:* This table projects the yield uplift (in basis points) from additional quantitative tightening beyond the 2025Q1 balance sheet, fixing the 2025Q1 portfolio and solving the nonlinear market-clearing system

$$\Delta \boldsymbol{y}_t = \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t^{\mathrm{CF}}\big) - \boldsymbol{g}\big(\boldsymbol{x}_t,\, \boldsymbol{\beta},\, \boldsymbol{\varepsilon}_t,\, \boldsymbol{A}_t,\, \bar{\boldsymbol{S}}_t\big),$$

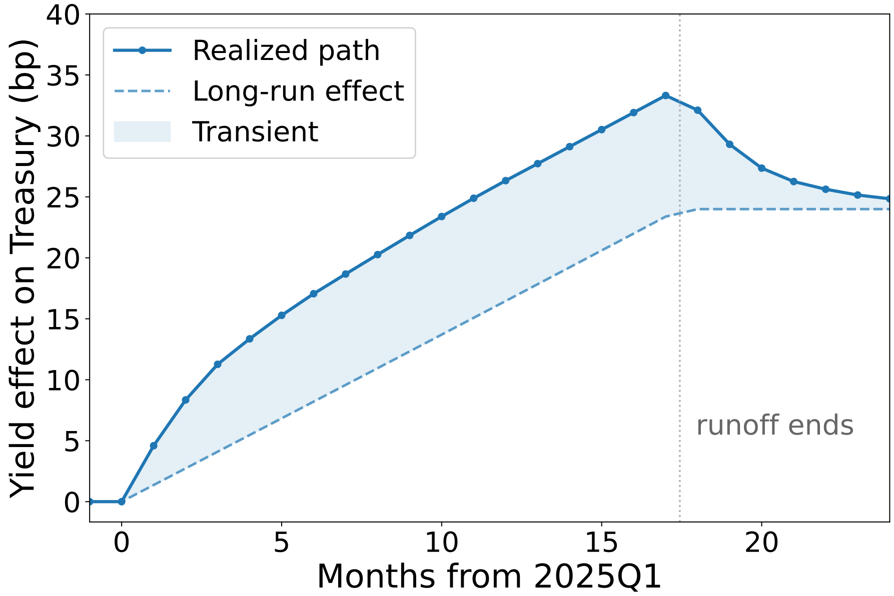
where $\bar{\boldsymbol{S}}_t^{\mathrm{CF}} = \bar{\boldsymbol{S}}_t + \boldsymbol{\Delta}\mathrm{SOMA}_t$ is the baseline float after the additional runoff, and $\boldsymbol{g}(\cdot)$ is the market-clearing inverse demand, which clears each market against the supply schedule of Equation (21). $\boldsymbol{\Delta}$SOMA is the additional Fed runoff beyond 2025Q1 (shown as a negative balance-sheet change), allocated between Treasuries and agency MBS at the 2024 average runoff pace; dashes indicate assets the Fed does not hold. Bond supply responds to the counterfactual yield through net-issuance elasticities estimated from the Flow of Funds (Section 5.2), accumulated over the projected runoff horizon (runoff amount divided by the 2024 pace). Corporate and Municipal entries capture cross-asset portfolio rebalancing only. The impacts are long-run equilibrium, reached once every sector completes rebalancing.

flows arrive and banks, insurers, and pension funds absorb their share of the released supply. Cumulating these vintages, Treasury yields rise by about two basis points per month while the runoff proceeds, peak at about 33 basis points when it ends, and settle at 24 basis points for the baseline \$1 trillion, 9 for the mild, and 39 for the aggressive scenario. The projection says most of the yield effect of incoming QT arrives *during* the runoff, roughly in proportion to the stock withdrawn, and that little further pressure follows once it stops. The gradual pace limits the transient: with the release spread over seventeen months, at most the most recent increments are in their overshoot phase at any time, and the path peaks about ten basis points above its long-run component before slow capital closes the gap. The same \$1 trillion released in a single quarter would instead move yields to more than three times their long-run level before slow capital arrives: the gradual pace is what spares the market the full dislocation. The impact overshoot and its decay are the price-impact counterpart of the horizon duality van der Beck (2026) documents for equities, and the adjustment dynamics of Greenwood et al. (2018), in which a supply shock is absorbed first by local specialists and only gradually by cross-market capital: a shock's initial price impact exceeds its long-run level and decays as slow capital arrives.

# 6 Conclusion

The runoff is ongoing, and the framework here is built for the questions it will keep raising: how far yields rise as the balance sheet shrinks, who absorbs the released supply, and how the pace shapes the transient. Three caveats bound the answers. The Flow of Funds reports no within-class maturity detail, so the system prices substitution across asset classes rather than along the curve; the supply side is deliberately reduced-form; and the demand slopes are identified from local variation while the policy experiments shift supply well outside it. Within those bounds, the projections say the remaining tightening is a matter of basis points rather than of market capacity, and that the gradual pace is what keeps it so.

Figure 5: **Projected Yield Path of Incoming QT**




*Note:* Projected yield effects of further runoff from the 2025Q1 portfolio at the 2024 pace (\$172 billion per quarter, rendered monthly at \$57 billion per month), split between Treasuries and agency MBS at the QT2 ratio. Each quarter's incremental runoff enters at its impact effect and converges to its long-run effect along the adjustment path estimated from the QT2 episode (the equilibrium re-solved horizon by horizon as the slow sectors travel from $\beta^{(0)}$ to $\beta^{(4)}$); the long-run component (dashed) accumulates the equilibrium effects of the runoff to date, with the supply response of Section 5.2 building over the elapsed horizon. The figure shows the Treasury path under the baseline \$1 trillion scenario at the actual pace, together with its long-run component (dashed); the shaded gap between them is the transient carried by the sectors still mid-adjustment, which peaks when the runoff ends and dissipates as slow capital arrives. The vertical dotted line marks the end of the runoff.

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# A A BLP Interpretation of Sector-Level Asset Demand

In this appendix, I introduce a within-sector investor-level utility representation that provides a BLP-style interpretation of the demand estimation in the paper. Intuitively, each sector aggregates many underlying institutions (e.g., the U.S. banking sector includes large banks such as JPMorgan Chase and Bank of America), but the empirical analysis uses sector-level portfolios rather than institution-level holdings. For each sector $i$ and within-sector investor $h$, the indirect utility from holding asset $n$ at time $t$ is

$$u_{hit}(n) = \beta_i \, y_t(n) + \gamma_i' x_t(n) + \xi_t(n) + v_{hit}(n), \tag{22}$$

while the outside asset $n = 0$ yields

$$u_{hit}(0) = \xi_t(0) + v_{hit}(0). \tag{23}$$

Here, $\beta_i$ and $\gamma_i$ denote sector-specific preference parameters that are common to all investors $h$ in sector $i$. The term $\xi_t(n)$ captures asset–time fixed effects that are common across sectors and investors, such as unobserved asset characteristics, aggregate risk premium, or market-wide valuation components. Finally, $v_{hit}(n)$ are i.i.d. Type-I extreme value shocks that generate idiosyncratic within-sector heterogeneity. Although random coefficient specifications are standard in the BLP literature, I restrict within-sector heterogeneity to idiosyncratic shocks because sector-level data do not contain sector-specific price variation required to identify a distribution of random coefficients. Put differently, each sector is analogous to a market in the standard BLP framework. However, all sectors observe the same asset prices in financial markets, which by construction provides insufficient price variation to identify within-sector heterogeneity.

Given the Type-I extreme value errors, the portfolio weight of sector $i$ on asset $n$ at time

$t$ is

$$w_{it}(n) = \frac{\exp\{u_{iht}(n)\}}{\sum_{k=0}^{N} \exp\{u_{iht}(k)\}}. \tag{24}$$

Normalize the outside option by setting $\xi_t(0) = 0$ and subtract $u_{iht}(0)$ from all utilities. Under this normalization, the portfolio weight can be written as in Equations (1) and (2):

$$w_{it}(n) = \frac{\delta_{it}(n)}{1 + \sum_{k=1}^{N} \delta_{it}(k)}.$$

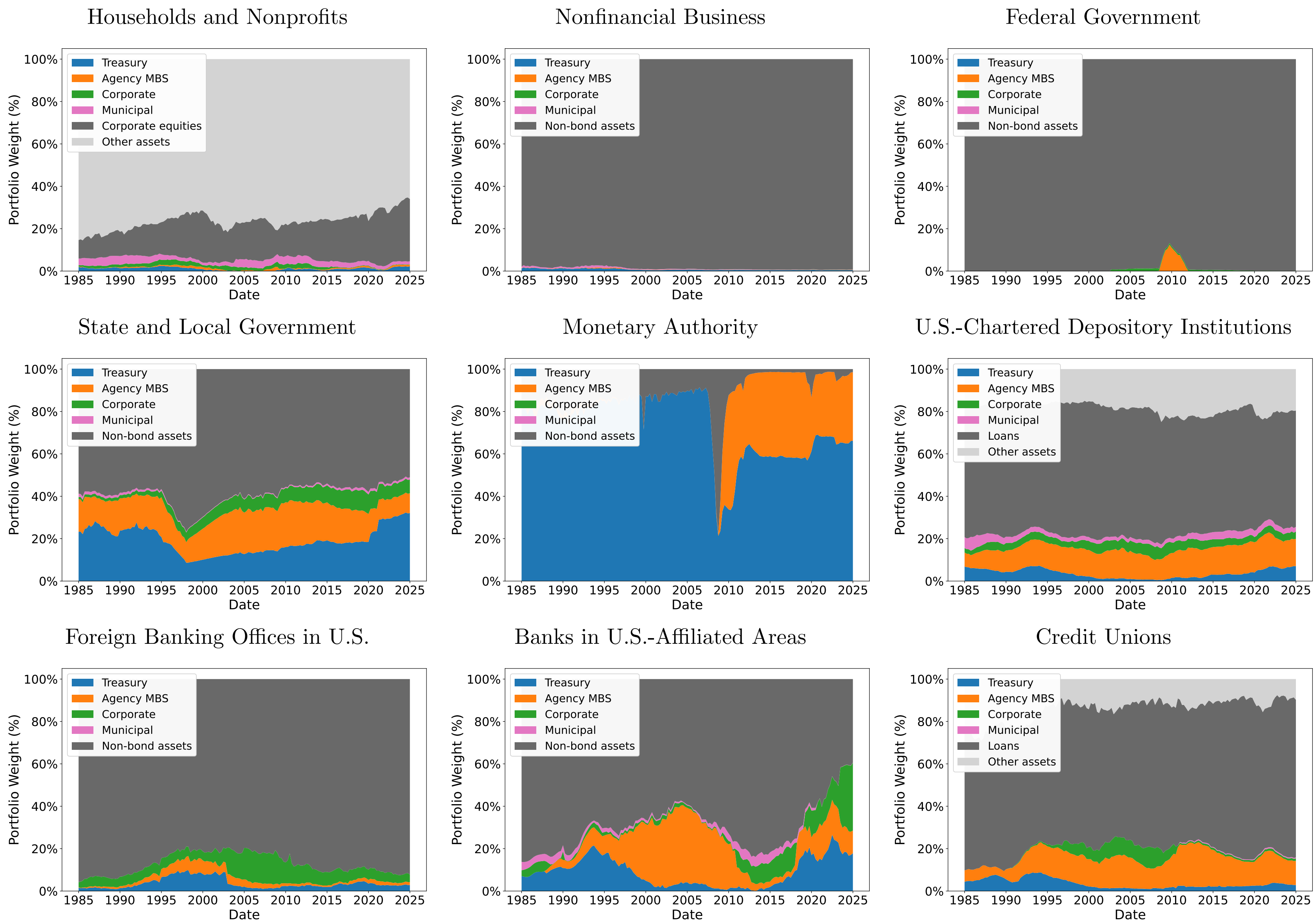
Here,

$$\delta_{it}(n) := \exp\Big\{\beta_i y_t(n) + \gamma_i' x_t(n) + \xi_t(n)\Big\}\, \epsilon_{it}(n),$$

where $\epsilon_{it}(n) := \exp\{v_{iht}(n) - v_{iht}(0)\}$.

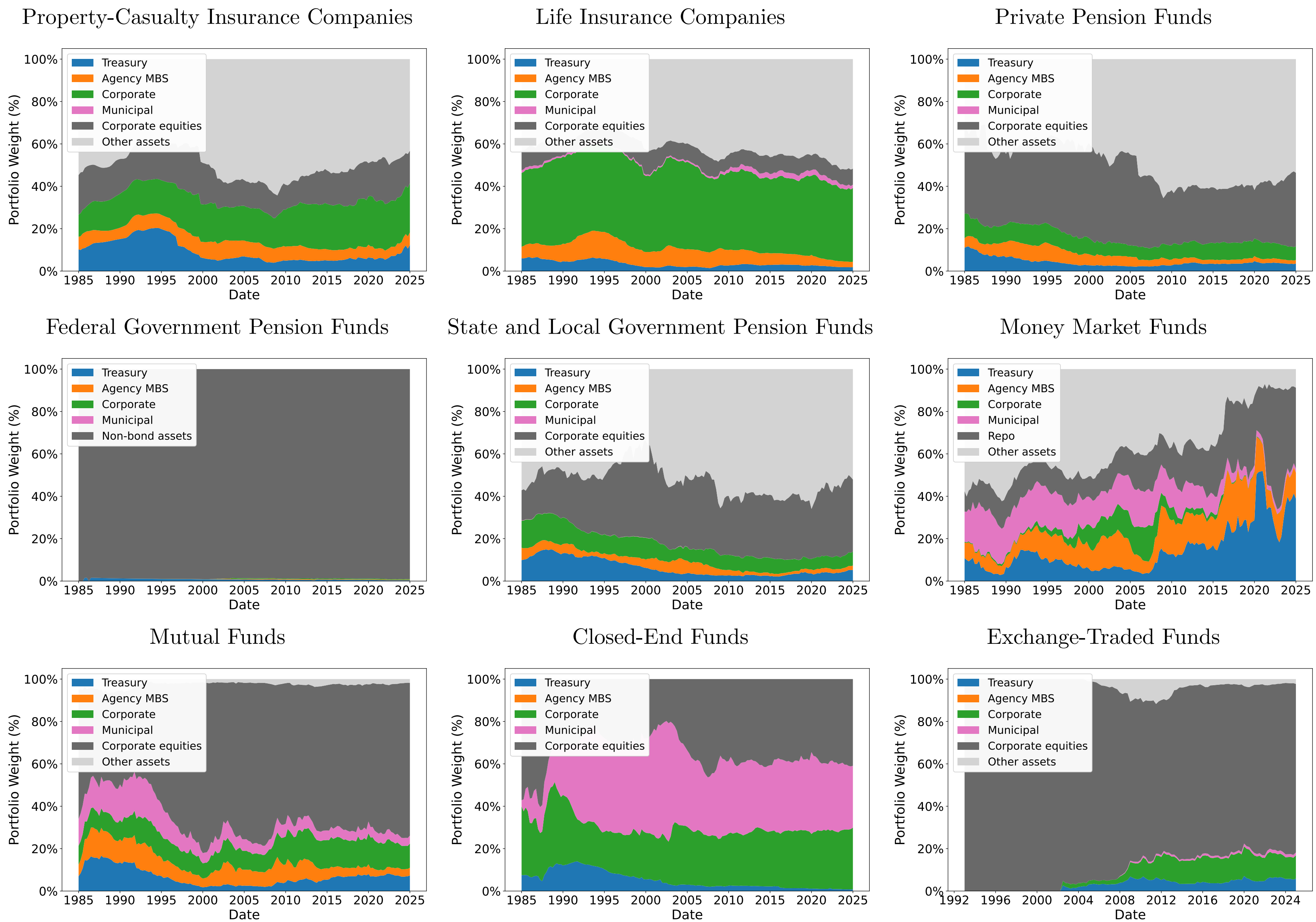
# B Figures

Figure A.6: **Portfolio Shares by Sector (continued)**




*Notes:* Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.6: **Portfolio Shares by Sector (continued)**

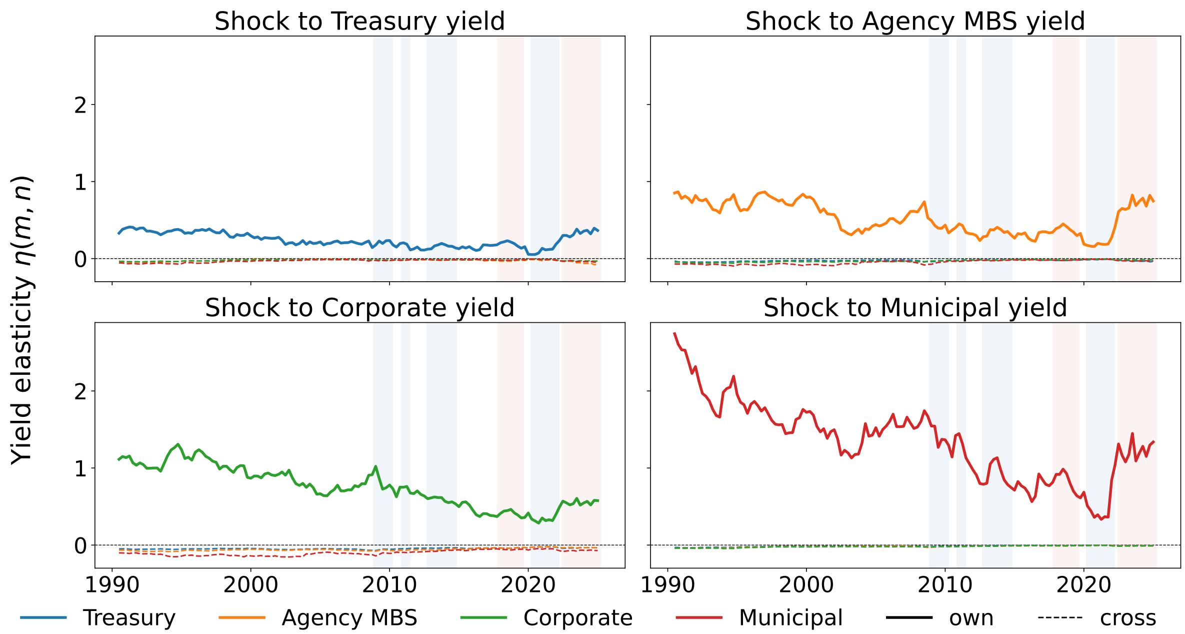



*Notes:* Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.6: **Portfolio Shares by Sector (continued)**


*Notes:* Stacked shares of total financial assets. Dark gray is the sector's own outside asset (Table 1); light gray is the remainder of the balance sheet, outside the demand system. For sectors whose outside asset is the total non-bond residual, the light-gray band is absent.

Figure A.7: **Own- and Cross-Yield Demand Elasticities**



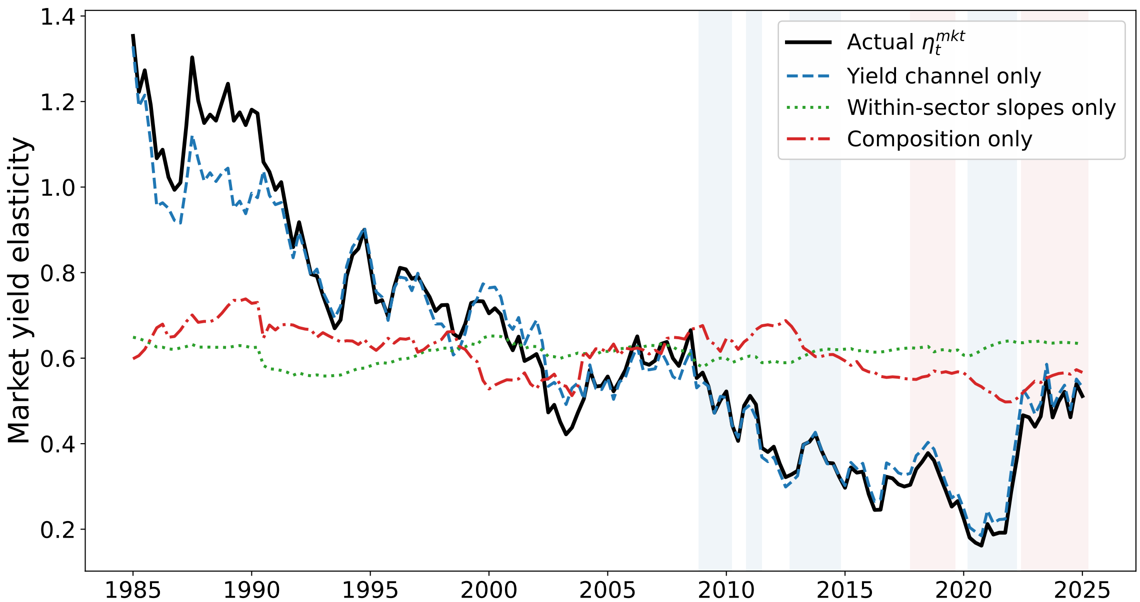

*Note:* Market-level yield elasticities $\eta_t^{\text{mkt}}(m,n) = \partial \log B_t(n)/\partial \log y_t(m)$, the holdings-weighted aggregate of the sector elasticities in Equation (8) at the counterfactual demand slopes ($\beta^{(4)}$ for every sector): each panel fixes the shocked yield $m$ and plots the own-response ($n = m$, solid) and the three cross-responses ($n \neq m$, dashed). Blue bands mark the four QE episodes, red bands the two QT episodes.

## B.1 Decomposing the Time Variation of the Market Elasticity

The market elasticity factors as $\eta_t^{mkt} = \bar{y}_t \sum_i \alpha_{it}\, \beta_i\, (1 - w_{it}^{\text{bond}})$, and the demand slopes $\beta_i$ are time-invariant, so its movement has exactly three sources: the yield level, the sectors' own bond shares (the $1-w$ term), and the composition of holdings $\alpha_{it}$. Figure A.8 plots the actual series against three counterfactuals that let one component move while holding the others at their sample means. The yield channel accounts for 93.8 percent of the quarterly variation in $\Delta \log \eta_t^{mkt}$; the within-sector share channel contributes 3.7 percent, and composition 2.6. In levels, the fall from 1.15 in 1985–1990 to 0.33 in the QE era decomposes into $-0.72$ from the yield level, with composition contributing $-0.10$ and within-sector shares essentially nothing, and the recovery to 0.46 after 2022 is again the yield level. The yield-free slope component is essentially flat over the sample, drifting from 0.128 to 0.106: the market's structural elasticity barely moved, and the QE-era decline in the elasticity is the yield level, not a change in who holds the market or how much they hold. The decomposition involves only own-elasticities, so it does not rest on the logit's cross-substitution structure.

# C Tables

Figure A.8: **Decomposition of the Market Demand Elasticity**




*Note:* The market elasticity $\eta_t^{mkt} = \bar{y}_t \sum_i \alpha_{it}\beta_i(1 - w_{it}^{\text{bond}})$ (black) against three one-channel counterfactuals, each letting a single component vary while the others are held at sample means: the yield level $\bar{y}_t$ (dashed), the within-sector slopes $\beta_i(1 - w_{it})$ at fixed composition (dotted), and the composition weights $\alpha_{it}$ at fixed slopes (dash-dotted). Sector-type level, with the counterfactual demand slopes ($\beta^{(4)}$). Blue bands mark the four QE episodes, red bands the two QT episodes.

Table A.9: **Data Sources for Sector-Level Portfolio Holdings**

| Type | Sector | Source |
|---|---|---|
| Banks | U.S.-Chartered Depositories | L.111 |
| | Foreign Banking Offices | L.112 |
| | Credit Unions | L.114 |
| | Banks in Affiliated Areas | L.113 |
| Insurers | Life Insurers | L.116 |
| | Property-Casualty Insurers | L.115 |
| Pension Funds | Private Pension Funds | L.118 |
| | State & Local Govt. Retirement Funds | L.120 |
| | Federal Govt. Retirement Funds | L.119 |
| Mutual Funds | Mutual Funds | L.122 |
| | Exchange-Traded Funds | L.124 |
| | Money Market Funds | L.121 |
| | Closed-End Funds | L.123 |
| Arbitrageurs | Brokers and Dealers | L.130 |
| | Hedge Funds | EFA |
| | mREITs | L.129 |
| Foreign | Rest of the World | L.133 |
| | Official / Private (post-2010) | CSLT |
| Household Residual | Household Residual & Nonprofits | L.101 |
| Other | Nonfinancial Business | L.102 |
| | Government-Sponsored Enterprises | L.125 |
| | Holding Companies | L.131 |
| | State & Local Governments | L.107 |
| | Federal Government | L.106 |
| | Agency/GSE Mortgage Pools | L.126 |
| | Finance Companies | L.128 |
| | ABS Issuers | L.127 |
| | Other Financial Business | L.132 |

*Notes:* Flow-of-Funds table codes (L. series) refer to the Financial Accounts of the United States (Federal Reserve Z.1). In the post-2010 refinement, foreign holdings are split into Foreign Official and Foreign Private using the Federal Reserve's CSLT estimates (Bertaut–Judson methodology; McCallum et al., 2026), since the Flow of Funds does not disaggregate foreign holdings by holder type. Hedge fund holdings are from the Federal Reserve's Enhanced Financial Accounts (EFA), built from SEC Form PF filings.

Table A.10: **Data Sources for Asset Yields and Characteristics**

| Asset | Variable | Source |
|---|---|---|
| Treasury | Yield | 10-year constant-maturity yield, DGS10, FRED |
| | Time to maturity | YAS_MTY_YEARS, Bloomberg (LUATTRUU) |
| | Modified duration | YAS_RISK, Bloomberg (LUATTRUU) |
| | Liquidity | Author's computation from CRSP |
| | Issuance size | FL893061105, Z.1 Financial Accounts (L.210) |
| Agency MBS | Yield | Bloomberg US MBS index yield (LUMSTRUU) |
| | Time to maturity | WAL_TO_MATURITY, Bloomberg (LUMSTRUU) |
| | Modified duration | YAS_RISK, Bloomberg (LUMSTRUU) |
| | Liquidity | Author's computation from TRACE |
| | Issuance size | FL893061705, Z.1 Financial Accounts (L.211) |
| Corporate | Yield | Moody's seasoned Baa corporate yield, DBAA, FRED |
| | Time to maturity | YAS_MTY_YEARS, Bloomberg (LUACTRUU) |
| | Modified duration | YAS_RISK, Bloomberg (LUACTRUU) |
| | Liquidity | Author's computation from TRACE |
| | Issuance size | FL893063005, Z.1 Financial Accounts (L.213) |
| Municipal | Yield | Bloomberg Municipal Bond index yield (LMBITR) |
| | Time to maturity | YAS_MTY_YEARS, Bloomberg (LMBITR) |
| | Modified duration | YAS_RISK, Bloomberg (LMBITR) |
| | Liquidity | Author's computation from MSRB RTRS |
| | Issuance size | FL893062005, Z.1 Financial Accounts (L.212) |

*Notes:* All yields are end-of-quarter. Maturity and duration are computed from the Bloomberg total-return index of each asset class (index tickers in parentheses); all indexes are unhedged and denominated in USD. The liquidity measures enter only the post-2010 robustness specification (Appendix D.4).

Table A.11: **Treasury Exogeneity: Three Instrument Families**

| Instrument set | $\beta$ | (s.e.) | 1st-stage $F$ | Sargan $J$ ($p$) |
|---|---|---|---|---|
| Granular $\Gamma$ (baseline) | +0.015 | (0.094) | 12.8 | — |
| Foreign shock only | +0.088 | (0.095) | 5.3 | — |
| $\Gamma$ + foreign (overid) | +0.030 | (0.087) | 6.9 | 0.38 |
| MP surprise (JK, 2008–) | −0.048 | (0.117) | 15.1 | — |
| $\Gamma$ + MP surprise (2008–) | −0.012 | (0.102) | 15.1 | 0.06 |

*Notes:* Pooled Treasury demand slope (duration-adjusted quarterly share changes on the yield change, domestic sectors only) under alternative instruments on 1985Q1–2025Q1. The foreign shock is the rest of the world's idiosyncratic Treasury-holding growth, the residual from a common-factor projection; foreign Treasury demand is reserve-driven and hence orthogonal to U.S. domestic demand shocks. The MP-surprise rows use the quarterly Jarociński and Karadi (2020) median monetary-policy shock (information shocks removed), on the post-2008 sample where it is relevant. Agreement across the three instrument families, and the Sargan $J$ tests of the joint specifications, corroborate the granular instrument's exclusion restriction. First-stage $F$ at the quarter level.

Table A.12: **QE4 Decomposition: Market-Functioning versus Stock-QE Phase**

| Phase | Fed Tsy ($B) | Tsy (bp) | MBS (bp) | Corp (bp) | $\lvert\Delta Y_{Tsy}\rvert/1\%$ |
|---|---|---|---|---|---|
| QE4 full | 3,511 | -183 | -107 | -9.6 | 8.8 |
| stock QE (from 2020Q2) | 1,240 | -76 | -61 | -4.5 | 10.3 |
| stock QE (from 2020Q3) | 1,000 | -62 | -56 | -4.1 | 10.6 |

*Notes:* QE4 (2019Q4–2021Q4) combined a market-functioning phase (2020Q1–Q2, $2.3 trillion of Treasury purchases in the COVID crash) with a later stock-QE phase. Rows re-base the counterfactual baseline to isolate the stock phase; the final column is the Treasury yield effect per 1 percent of float. The 2020H1 dislocation lies outside the stock-demand model, so the functioning phase is not reported as a stock effect. The phases are not additive, given the nonlinearity of the market-clearing map.

# D Specification Discussion

## D.1 Outside-Asset Assignment, Sector by Sector

The demand system measures each sector's bond demand relative to a single outside asset, the non-bond class the sector actually trades off against bonds (Table 1). This subsection records the rationale for each assignment, organized by outside asset.

**Loans: U.S.-chartered depositories and credit unions.** A depository's marginal portfolio decision is between its securities book and its loan book: deposits fund either, and the securities portfolio serves as the buffer that absorbs deposit inflows when loan demand is weak. The QE literature documents exactly this margin: banks receiving reserve inflows under QE expand lending and adjust securities holdings in tandem (Rodnyansky and Darmouni, 2017; Kandrac and Schlusche, 2021; Kurtzman et al., 2022). Measuring bank bond demand against total non-bond assets would instead lump in reserves and interbank claims, which move with monetary policy mechanically rather than through portfolio choice.

**Corporate equities: insurers, pension funds, mutual funds, and households.** For the long-term investors the relevant margin is the stock–bond allocation. Insurers and pension funds manage fixed-income portfolios against equity allocations under asset–liability and return targets; mutual funds, together with ETFs and closed-end funds, reallocate between bond and equity exposures through both manager choice and investor flows; and household financial portfolios trade off directly held bonds against directly held equities. This is the classic rebalancing margin of the portfolio-balance channel: when QE compresses bond yields, these investors are pushed toward equities, and the outside asset is defined to capture precisely that substitution.

**Repo: money market funds, broker-dealers, and hedge funds.** Money funds allocate between short government paper and repo daily; repo is their actual alternative at the margin, not equities they cannot hold. Dealers likewise manage securities inventory against matched-book and reverse-repo positions, so repo is the financing margin along which their bond holdings expand and contract. Hedge funds enter the data in Treasuries only, where the Enhanced Financial Accounts data are informative and where the cash–futures basis trade concentrates their holdings. That trade is repo-funded: the fund buys the cash Treasury and finances it in the repo market, so repo, not equities or other bonds, is the margin along which its Treasury book expands and contracts, the same financing margin as dealers and money funds. The assignment carries no structural weight, because hedge funds are not habitat investors: they are the model's arbitrageurs, whose Treasury positions respond to the cash–futures spread and to funding conditions rather than to the yield level, so a logit demand curve in the yield is not the right object for them. Using the EFA hedge-fund balance sheet, which reports the sector's repo assets and liabilities, a logit demand in the Treasury yield with repo as the outside asset gives a slope statistically indistinguishable from zero ($+0.03$, $t = 0.2$; $N = 49$, 2012Q4–2025Q1), the flow pattern expected of an arbitrage book rather than a demand curve. The margin is in fact degenerate: a repo-financed book expands the bond and its financing leg together, so the logit share ratio barely varies with the position, and the unit's own local-projection slope is unidentified (standard errors two orders of magnitude above the estimate); pooling the unit into the arbitrageur sector leaves the sector slope essentially unchanged. Accordingly, the carve-out serves measurement, correcting the household Treasury residual that hedge-fund positions dominate, and no hedge-fund demand parameter is estimated or reported. The main 26-sector panel nonetheless keeps hedge funds inside the household residual, because a full carve-out is infeasible: Form PF includes offshore-domiciled funds, whose holdings the Z.1 assigns to the rest of the world, so subtracting the full hedge-fund book from the residual double-counts the offshore share and produces negative household holdings. A domicile-adjusted carve-out is feasible, removing only the onshore share, at most $\varphi = 0.28$ of the hedge-fund book, the largest constant fraction consistent with non-negative carved holdings. Re-estimating the household local projections on the carved series (2012Q4–2025Q1) leaves the slopes essentially unchanged, $\beta^{(0)}$ moving from 0.52 to 0.58 and $\beta^{(4)}$ from $-1.28$ to $-1.38$: the impact slope is an artifact of the residual construction itself, not of the hedge funds inside it, and the household treatment in the counterfactual is robust to the carve.

**Non-bond U.S. securities: foreign investors.** The rest of the world's U.S. portfolio decision is between U.S. bonds and other U.S. securities, chiefly equities; the TIC-based holdings data are organized along exactly this line. Reserve managers, the dominant foreign Treasury holders, allocate between Treasuries, agencies, and other dollar assets, so the within-U.S. margin is the relevant one, not substitution back into home-currency assets.

**Total non-bond (residual): the remaining sectors.** For sectors without a single dominant alternative, foreign banking offices and banks in U.S.-affiliated areas (balance sheets dominated by interoffice claims), federal government retirement funds (holdings concentrated in special-issue Treasuries), mortgage REITs (levered MBS vehicles whose non-MBS assets are small), and the heterogeneous "Other" group, I use the residual of total financial assets net of the four bond classes. Imposing a specific margin on these balance sheets would assert substitution behavior the data cannot support; the residual is agnostic. With the exception of the mortgage REITs, whose elastic MBS demand is discussed above, these sectors hold small bond portfolios or enter the counterfactual with slopes near zero, so the choice has little quantitative weight.

**Does the assignment matter?** Table A.13 re-runs the full pipeline, local projections and the prospective-QT counterfactual, with every sector's outside asset replaced by the generic total-non-bond residual. The assignment matters where it should and not where it need not. Panel A shows that the sector-level slopes move substantially: the residual dilutes the margin of the long-term investors whose true alternative is a specific asset class, cutting the insurer slope from 0.09 to 0.05 and the mutual-fund slope from 0.29 to 0.22, while the arbitrageur slope rises from 0.55 to 0.70; the sectors whose outside asset is the residual by construction are unchanged. Panel B shows that the headline counterfactuals barely move: the baseline $1 trillion QT effect on Treasury yields shifts from 24.0 to 25.7 basis points, because market clearing depends on the holdings-weighted sum of the slopes and the sector-level shifts offset. The outside-asset assignment thus governs the composition of the response, which sector absorbs how much and how fast, while the aggregate yield effects are robust to it.

Table A.13: **Outside-Good Ablation: Sector-Specific versus Residual**

| *Panel A: Long-run demand slopes $\beta^{(4)}$ by sector* | | |
|---|---|---|
| Sector | Sector-specific | Residual |
| Banks | 0.07 | 0.08 |
| Insurers | 0.09 | 0.05 |
| Pension Funds | -0.07 | -0.11 |
| Mutual Funds | 0.29 | 0.22 |
| Arbitrageurs | 0.55 | 0.70 |
| Foreign Investors | 0.05 | 0.05 |
| Household Residual | 0.53 | 0.54 |
| Other | -0.03 | -0.03 |

| *Panel B: Prospective QT scenarios, yield effects (bp)* | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Sector-specific | | | | Residual | | | |
| Scenario | Tsy | MBS | Corp | Muni | Tsy | MBS | Corp | Muni |
| Mild ($400B) | 9.5 | 7.6 | 0.8 | 0.6 | 10.2 | 7.6 | 0.8 | 0.6 |
| Baseline ($1T) | 24.0 | 17.5 | 1.4 | 1.3 | 25.7 | 17.5 | 1.4 | 1.3 |
| Aggressive ($1.6T) | 38.7 | 26.1 | 1.7 | 2.0 | 41.5 | 26.1 | 1.7 | 1.9 |

*Notes:* Both columns re-estimate the full local-projection system and re-solve the prospective-QT scenarios of Section 5; the only change is the outside asset in the dependent variable $\ln(w_{it}(n)/w_{it}(0))$. "Sector-specific" assigns each sector the outside asset of Table 1; "Residual" assigns every sector total financial assets net of the four bond classes. Panel A reports the long-run slopes $\beta^{(4)}$ fed to the counterfactual engine (post-2010 window for mutual funds, arbitrageurs, and households, as in the main specification). Panel B reports the prospective-QT yield effects at the episode-end equilibrium.

## D.2 Splitting the Foreign Sector into Official and Private

The main specification treats foreign investors as one combined sector, entering the counterfactual as inelastic. The Federal Reserve's CSLT estimates permit a finer split into Foreign Official (reserve managers and central banks) and Foreign Private holdings from 2010 onward, and one may ask whether the counterfactual should use the two units separately, each with its own slope estimated on 2010–2025 as for the other post-2010 sectors. Table A.14 reports these estimates under the same specification and instrument menu as the main system.

The split is economically informative. Foreign Official demand is flat ($\beta = +0.09$, $t = 0.4$): reserve managers accumulate Treasuries for liquidity and exchange-rate objectives and do not respond to yields, the source of the foreign sector's inelasticity and of the exogeneity that makes foreign demand a useful corroborating instrument in Section 4. Foreign Private demand, by contrast, is significantly *negative* ($\beta = -0.39$, $t = -2.2$): private foreign holdings chase returns, rising when yields fall and prices appreciate, the same flow-driven pattern as mutual funds.

The split nonetheless does not enter the main counterfactual: a *negative* return-chasing coefficient describes momentum flows, not a demand curve along which the market can clear. Foreign investors hold large corporate bond positions, and imposing the negative private-foreign slope in market clearing reverses the corporate-yield spillover (QE4 would *raise* corporate yields by 39 basis points), a sign flip driven entirely by a coefficient that does not represent equilibrium substitution. The combined foreign sector, entering as inelastic, is the conservative treatment: it neither amplifies nor reverses the spillover and matches the Official component that dominates foreign Treasury holdings.

## D.3 Valuation Adjustment of Market-Value Holdings

The Financial Accounts report holdings at market value, so the observed change in a sector's log share moves for two distinct reasons: active purchases and sales, and the mechanical revaluation of existing positions when yields move. For a bond position of duration $D_t(n)$,

Table A.14: **Foreign Official versus Private Demand Slopes (2010–2025)**

| | Yield $\beta$ | Maturity | Duration | $N$ |
|---|---|---|---|---|
| Foreign Official | -0.09 | -0.03 | -0.01 | 244 |
| | (-0.6) | (0.06) | (0.07) | |
| Foreign Private | -0.60*** | 0.02 | 0.07 | 244 |
| | (-3.4) | (0.05) | (0.06) | |

*Notes:* Demand slopes for the two CSLT foreign components, estimated separately on 2010Q1–2025Q1 with the duration-adjusted first-difference specification and the instrument menu of the main system (granular instrument for Treasuries, SOMA purchases and the Drukker surprise for agency MBS, Lewbel instruments for corporate and municipal yields). Robust $t$-statistics in parentheses. The granular instrument excludes the entire rest-of-world sector, so it is valid for both foreign units. The counterfactual uses the combined foreign sector: the private component's negative slope reflects return-chasing flows rather than equilibrium substitution, and imposing it in market clearing reverses the corporate-yield spillover.

a yield change $\Delta y_t(n)$ (in percentage points) changes the position's market value by approximately $-D_t(n)\,\Delta y_t(n)/100$ even if the sector trades nothing. Left uncorrected, this mark-to-market term contaminates the differenced demand equation with a component that is mechanically negative in $\Delta y_t(n)$, producing a spurious downward-sloping response: sectors with long-duration portfolios would appear to sell when yields rise simply because their holdings are marked down. The bias is largest exactly where duration is longest, the long-term investors at the center of the analysis.

I therefore add the valuation term back and work with the active-rebalancing component of the share change,

$$\Delta q_{it}(n) \equiv \Delta \ln \frac{w_{it}(n)}{w_{it}(0)} + D_t(n)\,\frac{\Delta y_t(n)}{100}. \tag{25}$$

Eren et al. (2026) make the same correction with a constant eight-year duration (five and ten years in robustness); I use the observed, time-varying, asset-specific index duration $D_t(n)$ instead, which lets the adjustment reflect the shortening of MBS duration when rates fall (prepayment) and the drift in index duration over four decades. In the local projections, the adjustment is applied quarter by quarter and accumulated over the response window, so the left-hand side of the horizon regression measures cumulative active rebalancing between quarters $t-1$ and $t+h$. At long horizons the adjustment omits carry, the coupon and roll-down return that accrues over the window; because carry varies slowly, it is largely absorbed by the linear trend in the specification.

## D.4 Post-2010 Supply-Shock Specification

Before the extended sample and the granular instrument, the demand system was estimated on 2010Q1–2025Q1 with the supply-shock instruments of the prior literature, the Swanson (2021) monetary-policy surprises, SOMA purchases, and the maturity walls, on the finer 28-sector split that the post-2010 data support (Foreign Official/Private via CSLT, hedge funds via the EFA) and with the liquidity control available from TRACE and MSRB. Table A.15 reports these estimates. They serve two purposes: a robustness check on the main specification, and the only window in which the finest sector detail is observable. Because the yield $y_t(n)$ is common across sectors, the same instruments identify it everywhere and the first-stage relationship is shared; cross-sectional differences in the estimated yield sensitivities $\beta_i$ reflect genuine demand heterogeneity, not differences in the identifying variation. For the yield coefficient I report the 95% Anderson–Rubin confidence interval, which stays valid under the weak first stage (per-institution first-stage $F$ of roughly 5 to 10).

The yield coefficient is positive for most sectors, consistent with investors holding larger shares of higher-yielding assets, and its Anderson–Rubin interval excludes zero for 11 of the 28 sectors. Positive and significant demand appears for households ($\beta_i = 1.06$), both foreign sectors (Official 0.05, Private 0.06), life insurers (0.05), mutual funds (0.09), and the arbitrageur sectors, mREITs (0.17) and broker-dealers (0.10). The large household coefficient carries the same caveat as in the main specification (Section 4): the household sector is the Z.1 residual, which mechanically absorbs whatever the measured sectors do not, so its static coefficient reflects the residual construction rather than a behavioral elasticity, and the counterfactual does not use it. Most banks and pension funds show no significant yield response, consistent with allocations driven by regulatory requirements, investment mandates, or latent demand shocks rather than yield differentials alone; exchange-traded funds even load negatively ($-0.11$). Among the asset characteristics, households show the sharpest tilts, strongly toward large issues (issuance 12.8) and away from long-maturity, high-duration, and less-liquid bonds (maturity $-0.20$, duration $-0.22$, liquidity $-0.44$), and both foreign sectors share the preference for large issues (Official 0.62, Private 0.57). The fit ranges widely, with $R^2$ from 0.74 for hedge funds to near one for the foreign sectors. Every specification includes macro controls interacted with asset fixed effects and asset fixed effects; sector fixed effects do not apply, since I estimate each sector separately.

## D.5 Driscoll–Kraay Standard Errors

The baseline Newey–West standard errors correct the serial correlation that the overlapping local-projection windows induce, but not the correlation across assets and institutions within a quarter, which common macro shocks could generate despite the macro controls. Table A.16 therefore re-computes the standard errors of the sector-type long-run slopes $\beta^{(4)}$ with Driscoll–Kraay standard errors, which sum the scores within each quarter before applying the Bartlett kernel and so allow arbitrary cross-unit correlation within the quarter alongside the overlapping-window serial correlation; quarter-clustered standard errors, which capture the cross-unit correlation but drop the serial correlation, are shown for comparison. The point estimates are unchanged by construction. Every conclusion is intact: most slopes sharpen under Driscoll–Kraay, with the arbitrageur $t$-statistic rising from 1.8 to 2.5 and the household residual from 0.9 to 1.6, and the one estimate that widens, the insurer long-run slope, remains positive with $t = 1.6$.

## D.6 A Partial-Adjustment Benchmark

The local projections estimate each sector's adjustment path without imposing a functional form. The natural structural alternative is the partial-adjustment demand system of van der Beck (2026), in which each sector closes a constant fraction $\lambda_i$ of the gap between its current

Table A.15: **Estimated Demand Parameters by Sector (Post-2010 Supply-Shock Specification)**

| | Demand coefficients | | | | | | |
|---|---|---|---|---|---|---|---|
| Sector | Yield | Maturity | Issuance | Duration | Liquidity | $N$ | $R^2$ |
| (a) Banks | | | | | | | |
| Banks in Affiliated Areas | 0.02 | 0.23** | 0.47 | -0.21*** | 0.16** | 235 | 0.774 |
| | [-0.80, 0.90] | (0.09) | (2.17) | (0.08) | (0.08) | | |
| Credit Unions | 0.09 | 0.50 | 4.69*** | -0.72* | 0.16*** | 235 | 0.780 |
| | [-0.72, 1.08] | (0.31) | (1.75) | (0.38) | (0.05) | | |
| Foreign Banking Offices | -0.11 | -0.06** | -1.05** | -0.00 | 0.05*** | 235 | 0.999 |
| | [-0.22, -0.00] | (0.02) | (0.49) | (0.03) | (0.02) | | |
| U.S.-Chartered Depositories | -0.04 | -0.01 | 0.76** | -0.04** | 0.03** | 235 | 0.979 |
| | [-0.14, 0.04] | (0.02) | (0.36) | (0.02) | (0.01) | | |
| (b) Insurers | | | | | | | |
| Life Insurers | 0.05** | 0.02 | 0.86** | -0.06*** | 0.01 | 235 | 0.994 |
| | [0.08, 0.12] | (0.02) | (0.37) | (0.02) | (0.01) | | |
| Property-Casualty Insurers | 0.08 | 0.04* | 0.02 | -0.10*** | -0.05*** | 235 | 1.000 |
| | [-0.02, 0.16] | (0.03) | (0.39) | (0.02) | (0.01) | | |
| (c) Pension Funds | | | | | | | |
| Federal Government Funds | -0.06** | -0.03** | -0.28 | 0.03** | -0.02*** | 235 | 1.000 |
| | [-0.08, -0.04] | (0.01) | (0.21) | (0.01) | (0.01) | | |
| Private Pension Funds | -0.03 | -0.03 | 0.33 | 0.03** | -0.01 | 235 | 1.000 |
| | empty | (0.02) | (0.43) | (0.01) | (0.01) | | |
| State & Local Government Funds | 0.00 | -0.24*** | -0.39 | 0.20** | 0.09*** | 235 | 0.973 |
| | [-0.72, 0.66] | (0.08) | (0.83) | (0.10) | (0.03) | | |
| (d) Mutual Funds | | | | | | | |
| Closed-End Funds | -0.01 | -0.01 | -0.07 | -0.01 | -0.01** | 235 | 1.000 |
| | [-0.04, 0.02] | (0.01) | (0.23) | (0.01) | (0.01) | | |
| Exchange-Traded Funds | -0.11** | 0.07*** | -1.54*** | -0.05 | -0.05*** | 235 | 0.999 |
| | [-0.20, -0.08] | (0.03) | (0.50) | (0.03) | (0.02) | | |
| Money Market Funds | 0.16 | -0.02 | 2.68*** | 0.00 | 0.08*** | 235 | 0.978 |
| | [-0.10, 0.46] | (0.03) | (0.55) | (0.03) | (0.03) | | |
| Mutual Funds | 0.09** | 0.07** | 1.17** | -0.07*** | 0.01 | 235 | 0.941 |
| | [0.06, 0.20] | (0.03) | (0.53) | (0.02) | (0.02) | | |
| (e) Arbitrageurs | | | | | | | |
| Hedge Funds | -0.20 | 0.24 | 0.04 | -0.66*** | -0.02 | 49 | 0.743 |
| | [-0.44, 0.04] | (0.15) | (0.36) | (0.11) | (0.02) | | |
| mREITs | 0.17** | 0.02 | 0.74 | -0.17*** | -0.10*** | 235 | 0.997 |
| | [0.10, 0.38] | (0.05) | (0.89) | (0.05) | (0.04) | | |
| Brokers and Dealers | 0.10** | -0.04 | 2.28*** | -0.02 | 0.02 | 235 | 0.999 |
| | [0.04, 0.20] | (0.03) | (0.70) | (0.02) | (0.02) | | |

Table A.15: **Estimated Demand Parameters by Sector (Post-2010)** (continued)

| Sector | Demand coefficients | | | | | $N$ | $R^2$ |
|---|---|---|---|---|---|---|---|
| | Yield | Maturity | Issuance | Duration | Liquidity | | |
| (f) Foreign | | | | | | | |
| Foreign Official | 0.05** | -0.04 | 0.62 | -0.01 | -0.01 | 235 | 1.000 |
| | [0.04, 0.18] | (0.03) | (0.48) | (0.02) | (0.02) | | |
| Foreign Private | 0.06** | 0.01 | 0.57 | -0.06*** | -0.02** | 235 | 1.000 |
| | [0.06, 0.10] | (0.02) | (0.37) | (0.01) | (0.01) | | |
| (g) Households | | | | | | | |
| Households & Nonprofits | 1.06** | -0.20** | 12.76*** | -0.22*** | -0.44*** | 175 | 0.910 |
| | [0.78, 1.60] | (0.09) | (1.58) | (0.07) | (0.08) | | |
| (h) Other | | | | | | | |
| Agency/GSE Mortgage Pools | 0.00 | 0.00 | 0.00 | -0.00 | -0.00 | 235 | — |
| | [-0.00, -0.00] | (0.00) | (0.00) | (0.00) | (0.00) | | |
| Federal Government | 1.43** | 0.08 | 14.38** | -0.29* | -0.38** | 235 | 0.913 |
| | [1.12, 2.68] | (0.24) | (5.85) | (0.17) | (0.18) | | |
| Finance Companies | -0.08 | 0.02* | -0.61** | 0.05*** | -0.07*** | 235 | 1.000 |
| | [-0.16, -0.00] | (0.01) | (0.25) | (0.01) | (0.01) | | |
| GSEs | 0.12 | -0.03 | 1.26** | -0.02 | 0.08*** | 235 | 0.999 |
| | empty | (0.02) | (0.56) | (0.02) | (0.02) | | |
| Holding Companies | -0.10 | -0.05 | -1.38 | 0.03 | 0.01 | 235 | 0.996 |
| | empty | (0.04) | (0.97) | (0.03) | (0.03) | | |
| ABS Issuers | 0.39** | 0.00 | 3.71** | 0.13 | 0.03 | 235 | 0.988 |
| | [0.22, 0.80] | (0.10) | (1.60) | (0.12) | (0.05) | | |
| Nonfinancial Business | 0.01 | 0.05* | 0.35 | -0.04* | -0.01 | 235 | 0.986 |
| | [-0.12, 0.12] | (0.03) | (0.49) | (0.02) | (0.02) | | |
| Other Financial Business | 0.16 | -0.10*** | 1.31* | 0.01 | -0.05 | 235 | 0.998 |
| | empty | (0.03) | (0.79) | (0.03) | (0.03) | | |
| State & Local Governments | 0.08** | 0.02* | 1.05*** | -0.04*** | -0.07*** | 235 | 0.997 |
| | [0.02, 0.18] | (0.01) | (0.22) | (0.01) | (0.01) | | |

*Notes:* I estimate the equation below separately for each sector on 2010Q1–2025Q1, instrumenting the yield with the supply-shock menu of the prior literature (monetary-policy surprises, SOMA purchases, maturity walls). This is the *level* specification of that literature, with asset fixed effects, reported as a specification check; the main results are instead estimated in first differences with local projections, Equation (7), on the 1985–2025 sample. The yield coefficient $\beta_i$ is the semi-elasticity of the log portfolio share relative to the outside asset with respect to a one-percentage-point change in yield; the characteristic coefficients $\gamma_i$ measure sensitivities to maturity, issuance, duration, and liquidity. For the yield coefficient, brackets report the 95% Anderson–Rubin confidence interval, which stays valid under weak instruments; stars mark intervals that exclude zero. The Anderson–Rubin set is "empty" when the data reject the overidentifying restrictions and "unbounded" when the instruments fail to identify $\beta_i$. For the characteristic coefficients, parentheses report robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

$$\ln \frac{w_{it}(n)}{w_{it}(0)} = \beta_i y_t(n) + \gamma_i' x_t(n) + \alpha_n + \nu_n' W_t + \varepsilon_{it}(n).$$

Table A.16: **Long-Run Demand Slopes under Alternative Standard Errors**

| Sector | $\beta^{(4)}$ | HAC | Clustered | Driscoll–Kraay | Window |
|---|---|---|---|---|---|
| Banks | 0.068 | (0.179) | (0.101) | (0.102) | 1985–2025 |
| Insurers | 0.095 | (0.047) | (0.060) | (0.060) | 1985–2025 |
| Pension Funds | -0.070 | (0.073) | (0.055) | (0.051) | 1985–2025 |
| Mutual Funds | 0.290 | (0.264) | (0.189) | (0.191) | 2010–2025 |
| Arbitrageurs | 0.550 | (0.299) | (0.233) | (0.223) | 2010–2025 |
| Foreign | 0.047 | (0.046) | (0.049) | (0.048) | 1985–2025 |
| Household Residual | 0.534 | (0.577) | (0.450) | (0.326) | 2010–2025 |
| Other | -0.028 | (0.082) | (0.088) | (0.102) | 1985–2025 |

*Notes:* The sector-type long-run slope $\beta^{(4)}$ from the pooled local projection, Equation (7), with three standard errors in parentheses: the baseline Newey–West (Bartlett kernel, bandwidth 4); clustered by quarter; and Driscoll–Kraay (scores summed within each quarter, Bartlett kernel with four lags), which allows both cross-unit correlation within the quarter and the moving-average serial correlation of the overlapping windows. Point estimates are identical across columns by construction. Windows follow the display rule of Table 4: mutual funds, the arbitrageurs, and the household residual on 2010Q1–2025Q1, all other sectors on 1985Q1–2025Q1.

allocation and the static logit target each quarter, so that the long-run slope is recovered from one-quarter inertia as $\beta_{LR} = \theta_y/\lambda_i$. As a cross-check, I estimate this model on the same panel, windows, and instruments as the local projections, in error-correction form: the duration-adjusted rebalancing $\Delta q_{it}(n)$ is regressed on the lagged level $q_{i,t-1}(n)$, the lagged yield, and the instrumented yield change, with unit-by-asset fixed effects, asset trends, and the macro controls; $\lambda_i$ is the negative of the coefficient on the lagged level.

Table A.17 reports the results, and they cross-validate the design in one dimension while cautioning against it in the other. The long-run slopes agree: for six of the eight sector types the partial-adjustment $\beta_{LR}$ lies within roughly one standard error of the local-projection $\beta^{(4)}$, with the same cross-sector ordering, elastic mutual funds and arbitrageurs against inelastic long-term investors. The exceptions are instructive: for insurers, the model extrapolates the slow climb into a long-run target well above the directly estimated plateau, and for households it loads the mean reversion of the Z.1 residual onto a large $\lambda$, the same artifact discussed in Section 4. The adjustment speeds, however, are not credible: the implied half-lives run from under three quarters to over twelve years, far slower than the directly estimated paths, which are flat within a year for every sector (Figure 2). This is the signature of persistent latent demand: in a lagged-level model, any persistence in $\xi_{it}(n)$ loads onto the single inertia parameter, conflating slow rebalancing with slowly moving preferences. The local projections separate the two, which is why the paper reads both the long-run slopes and the speeds from the projections and treats the partial-adjustment model as a benchmark only.

Table A.17: **Partial-Adjustment Benchmark versus Local Projections**

| | Adjustment | Half-life | Long-run slope | | Impact | |
|---|---|---|---|---|---|---|
| Sector type | $\lambda$ | (quarters) | PA $\beta_{LR}$ | LP $\beta^{(4)}$ | LP $\beta^{(0)}$ | Sample |
| Banks | +0.04*** | 15.2 | -0.00 | +0.07 | +0.03 | 1985–25 |
| | (0.02) | | (0.18) | | | |
| Insurers | +0.03*** | 23.8 | +0.55*** | +0.09 | +0.04 | 1985–25 |
| | (0.01) | | (0.18) | | | |
| Pension Funds | +0.11* | 6.0 | +0.03 | -0.07 | -0.02 | 1985–25 |
| | (0.06) | | (0.11) | | | |
| Mutual Funds | +0.04*** | 16.0 | +0.36** | +0.29 | -0.05 | 2010–25 |
| | (0.01) | | (0.17) | | | |
| Arbitrageurs | +0.06** | 11.5 | +0.27 | +0.55 | +0.21 | 2010–25 |
| | (0.02) | | (0.18) | | | |
| Foreign | +0.01 | 51.9 | -0.01 | +0.05 | +0.04 | 1985–25 |
| | (0.01) | | (0.28) | | | |
| Household Residual | +0.22*** | 2.8 | +0.34*** | +0.53 | +0.72 | 2010–25 |
| | (0.07) | | (0.12) | | | |
| Other | +0.02*** | 38.7 | +0.11 | -0.03 | +0.01 | 1985–25 |
| | (0.01) | | (0.28) | | | |

*Notes:* Partial-adjustment (PA) demand system in the spirit of van der Beck (2026), estimated per sector type in error-correction form on the same panel, estimation windows, and instrument sets as the local projections: $\Delta q_{it}(n) = \theta_{\Delta y}\,\Delta y_t(n) + \theta_y\, y_{t-1}(n) + \theta_q\, q_{i,t-1}(n) +$ controls $+\, u_{it}(n)$, with $\Delta y_t(n)$ instrumented, unit-by-asset fixed effects, asset-specific trends, and macro controls. The adjustment rate is $\lambda = -\theta_q$, the half-life is $\ln(0.5)/\ln(1-\lambda)$ in quarters, and the implied long-run slope is $\beta_{LR} = \theta_y/\lambda$, with delta-method standard errors in parentheses. The final two columns repeat the local-projection slopes of Table 4. Newey–West standard errors. $^{*}$, $^{**}$, and $^{***}$ denote $|t|$ above 1.64, 1.96, and 2.58.

## D.7 The OLS Benchmark

Table A.18 estimates the local projections of Section 4.2 by ordinary least squares, with the same panel, windows, controls, and trends but no instruments, next to the IV baseline. The comparison documents the direction of the simultaneity bias. OLS understates the long-run slope wherever demand is elastic, and the compression is most severe exactly there: the mutual-fund slope falls from 0.29 to 0.02, the arbitrageur slope from 0.55 to 0.08, and the household-residual slope from 0.53 to $-0.07$, while the inelastic sectors move far less. The pattern is what simultaneity predicts. A sector's latent demand raises prices and lowers yields, biasing the OLS slope downward, and the elastic sectors' rebalancing is precisely the demand that moves the market, so their bias is largest. Estimated by OLS, the demand system would find essentially no elastic capital in the bond market and overstate the yield effects of balance-sheet policy accordingly. The IV estimates recover the elastic sectors; since any residual weak-instrument bias runs toward OLS, the IV slopes remain lower bounds and the counterfactual yield effects conservative.

Table A.18: **OLS versus IV Local-Projection Slopes**

| Sector type | Impact $\beta^{(0)}$ OLS | Impact $\beta^{(0)}$ IV | Long-run $\beta^{(4)}$ OLS | Long-run $\beta^{(4)}$ IV | Sample |
|---|---|---|---|---|---|
| Banks | +0.06 | +0.03 | +0.07 | +0.07 | 1985–25 |
| | (0.01) | (0.05) | (0.04) | (0.18) | |
| Insurers | +0.04 | +0.04 | +0.06 | +0.09 | 1985–25 |
| | (0.01) | (0.02) | (0.01) | (0.05) | |
| Pension Funds | +0.03 | -0.02 | +0.03 | -0.07 | 1985–25 |
| | (0.03) | (0.05) | (0.03) | (0.07) | |
| Mutual Funds | +0.05 | -0.05 | +0.02 | +0.29 | 2010–25 |
| | (0.02) | (0.08) | (0.05) | (0.26) | |
| Arbitrageurs | +0.08 | +0.21 | +0.08 | +0.55 | 2010–25 |
| | (0.03) | (0.09) | (0.05) | (0.30) | |
| Foreign Investors | +0.05 | +0.04 | +0.07 | +0.05 | 1985–25 |
| | (0.01) | (0.02) | (0.01) | (0.05) | |
| Household Residual | -0.02 | +0.72 | -0.07 | +0.53 | 2010–25 |
| | (0.08) | (0.45) | (0.17) | (0.58) | |
| Other | +0.07 | +0.01 | +0.09 | -0.03 | 1985–25 |
| | (0.01) | (0.03) | (0.03) | (0.08) | |

*Notes:* Sector-type local-projection slopes at the impact ($h = 0$) and long-run ($h = 4$) horizons, estimated by OLS (no instruments) and by the baseline IV specification on the same panel, estimation windows, controls, and asset-specific trends. Newey–West standard errors in parentheses. The final column reports each sector type's estimation window, as in Table 4.